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Continuous Random Variables

A continuous random variable is a random variable that has only continuous values.
A random variable X with CDF $F_X(x)$ is said to be continuous if $F_X(x)$ is a continuous function for all x∈R.

More formally, given any interval $I = [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$, a random variable $X_I \sim \mathrm{U}(I) = \mathrm{U}[a, b]$ is called a "continuous uniform random variable" (CURV) if the probability that it takes a value in a subinterval depends only on the length of the subinterval. This implies that the probability of X_I falling in any subinterval $[c, d] \subseteq [a, b]$ is proportional to the length of the subinterval, that is, if $a \leq c \leq d \leq b$, one has

$$\Pr(X_I \in [c, d]) = \frac{d - c}{b - a}$$

We say that X is a continuous random variable if ∃ a non-negative function f(x)
∀ real X ∈ (-∞, ∞), having the property that for any set B of real #s,

$$P\{x \in B\} = \int_B f(x)\,dx$$

The function f(x) is called the probability density function of the rv X.

For a continuous random variable,

$$P(X = x) = 0 \quad \forall\ x. \qquad \int_x^x f(x) = 0$$

X must take a value in (-∞, ∞), so f(x) must satisfy

$$\int_{-\infty}^{\infty} f(x) = 1$$

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

Q. Suppose that X is a continuous rv whose PDF is given as

$$f(x) = \begin{cases} c(4x - 2x^2), & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

Calculate c. Also find the probability P(X>1).

$\longrightarrow \int_{-\infty}^{\infty} f(x)\,dx = 1$

$\Rightarrow 0 + \int_0^2 c(4x - 2x^2) + 0 = 1$

$\Rightarrow \left[4\frac{x^2}{2} - 2\frac{x^3}{3}\right]_{x=0}^{2} = \frac{1}{c}$

$\Rightarrow 2 \cdot 4 - \frac{2}{3} \cdot 8 = \frac{1}{c}$

$\Rightarrow 8 - \frac{16}{3} = \frac{1}{c}$

$\Rightarrow \frac{8}{3} = \frac{1}{c}$

$\Rightarrow c = \frac{3}{8}$

$$P(x > 1) = \int_1^{\infty} f(x)\,dx$$

$$= \int_1^2 c \cdot (4x - 2x^2)\,dx$$

$$= \frac{3}{8} \int_1^2 (4x - 2x^2)\,dx$$

$$= \frac{3}{8}\left[4\frac{x^2}{2} - 2\frac{x^3}{3}\right]_1^2$$

$$= \frac{3}{8}\left[2(3) - \frac{2}{3}(7)\right]$$

$$= \frac{3}{8}\left(6 - \frac{14}{3}\right)$$

$$= \frac{3}{8} \cdot \frac{4}{3} = \frac{1}{2}$$

Cumulative Distribution Function

The cumulative distribution function F is represented by

$$F(x) = P\{X \in (-\infty, x)\} = \int_{-\infty}^{a} f(x)\,dx$$

$$\frac{d}{dx} F(x) = f(x)$$

Q. $f(x) = \begin{cases} cx, & 2 \le x \le 10 \\ 0, & \text{otherwise} \end{cases}$ is the prob density function of X. Calculate c. Calculate the CDF of X.

→ $\int_2^{10} cx = 1$

$\left[\frac{x^2}{2}\right]_2^{10} = \frac{1}{c}$

$\frac{100-4}{2} = \frac{1}{c} = 48$

$c = \frac{1}{48}$

$f(x) = \begin{cases} x/48, & 2 \le x \le 10 \\ 0 \end{cases}$

$F(x) = \int_{-\infty}^{x} f(x)$

For $-\infty < x < 2$: $F(x) = 0$

For $2 \le x \le 10$: $F(x) = \int_{-\infty}^{x} f(x)\,dx$

$= \int_2^x f(x)\,dx = \frac{1}{48}\int_2^x x\,dx = \frac{1}{48}\left[\frac{x^2-4}{2}\right] = \frac{x^2-4}{96}$

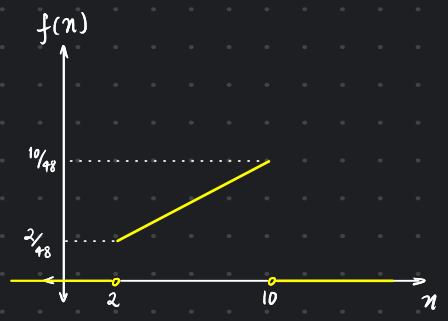
For $x > 10$: $F(x) = \int_{-\infty}^{x} f(x)\,dx = \int_{-\infty}^{10} f(x)\,dx + \int_{10}^{x} f(x)\,dx$

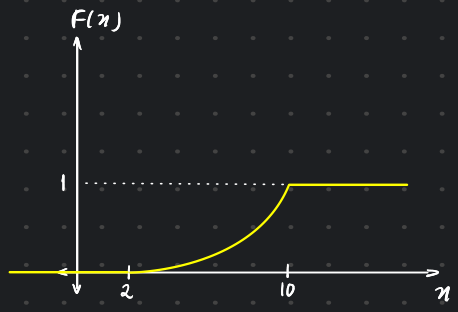
$= 1 + 0 = 1$

OBSERVE: $F(2) = 0$

$F(10) = 1$

$F(x > 10) = 1$





Find the prob $P(x^2 - 12x + 35 > 0)$

→ $x^2 - 12x + 35 > 0$

$(x-7)(x-5) > 0$

$x < 5$ or $x > 7$

$P(X \in (-\infty, 5) \cup (7, \infty))$

$= \int_{-\infty}^{5} f(x)\,dx + \int_{7}^{\infty} f(x)\,dx$

$= F(5) + 1 - F(7)$

$= \left[\frac{x^2-4}{96}\right]_{x=5} + 1 - \left[\frac{x^2-4}{96}\right]_{x=7}$

$= \frac{21}{96} + \frac{96}{96} - \frac{45}{96} = \frac{72}{96} = \frac{3}{4}$

Q. Let X be a continuous rv with the following PDF:

$$f(x) = \begin{cases} ce^{-x}, & x > 0 \\ 0, & \text{otherwise} \end{cases} \qquad c \text{ is a +ve constant}$$

a. Find c.

→ $\int_0^\infty c \cdot e^{-x}\, dx = 1 \qquad f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$

$\Rightarrow [-e^{-x}]_0^\infty = \frac{1}{c}$

$\Rightarrow e^{-0} - e^{-\infty} = \frac{1}{c}$

$\Rightarrow 1 - 0 = \frac{1}{c}$

$\Rightarrow c = 1$

b. Find the CDF of X.

→ $F(X) = \int_{-\infty}^{x} f(x)\, dx$

for $-\infty < x < 0$: $F(x) = \int_{-\infty}^{x} 0\, dx = 0$

for $0 < x < \infty$: $F(x) = \int_{-\infty}^{x} f(x)\, dx = \int_{-\infty}^{0} 0\, dx + \int_0^x e^{-x}\, dx = \left[-e^{-x}\right]_{x=0}^{x=x} = 1 - e^{-x}$

$$F(x) = \begin{cases} 1 - e^{-x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

c. Find P(1<X<5).

→ $P(1 < X < 5) = \int_1^5 e^{-x}\, dx = \left[-e^{-x}\right]_1^5 = \left[e^{-x}\right]_5^1 = e^{-1} - e^{-5} = 0.3611$

Example for joint PMF

Example for the discrete case:
Suppose that 3 balls are randomly selected from a bag containing 3 red, 4 white, and 5 blue balls. If we let X and Y denote the # of red and white balls taken out respectively, find the joint PMF of X and Y.

3 R, 4 W, 5 B → 3 balls $\binom{12}{3}$ total ways

$X : \{0, 1, 2, 3\}$

$Y : \{0, 1, 2, 3\}$

$P(X=0, Y=0) = \frac{{}^5C_3}{{}^{12}C_3} = \frac{10}{220} = P(0,0)$

$P(0,1) = \frac{{}^4C_1 \cdot {}^5C_2}{{}^{12}C_3} = \frac{40}{220} \qquad P(1,1) = \frac{{}^3C_1\, {}^4C_1\, {}^5C_1}{{}^{12}C_3} = \frac{60}{220} \quad \ldots$

TABLE 6.1: $P(X = i, Y = j)$

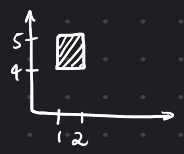
i \ j	0	1	2	3	Row sum $= P(X = i)$
0	$\frac{10}{220}$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$\frac{84}{220}$
1	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$\frac{108}{220}$
2	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$\frac{27}{220}$
3	$\frac{1}{220}$	0	0	0	$\frac{1}{220}$
Column sum $= P(Y = j)$	$\frac{56}{220}$	$\frac{112}{220}$	$\frac{48}{220}$	$\frac{4}{220}$	

Jointly distributed random variables

We say that X and Y are jointly continuous if there exists a function f(x, y) defined for all real x and y, having the property that for every set C of pairs of real #s,

$$P\{(X,Y) \in C\} = \iint_{x,y \in C} f(x,y)\,dx\,dy \qquad \begin{array}{l} x \sim (-\infty, \infty) \\ y \sim (-\infty, \infty) \end{array}$$

e.g. $C = \{1 \le x \le 2,\ 4 \le y \le 5\}$



The function f(X, Y) is called the joint probability density function of X and Y.
If A and B are any sets of real #s, then by defining $C \in \{(x, y) : x \in A, y \in B\}$,

$$P($$

Joint cumulative distribution function

$$F(a, b) = P\{X \in (-\infty, a]$$

Q. The joint density function of X and Y is given as

$$f(x,y) = \begin{cases} 2e^{-x}e^{-2y}, & 0<x<\infty,\ 0<y<\infty \\ 0, & \text{otherwise} \end{cases}$$

a. Calculate P{X>1, Y<1}.

$$P\{X>1, Y<1\} = \int_{-\infty}^{1}\left[\int_{1}^{\infty} 2e^{-x}e^{-2y}\,dx\right]dy$$

$$= \int_{-\infty}^{1} 2e^{-2y}\left[-e^{-x}\right]_{x=1}^{\infty} dy = \int_{-\infty}^{1} 2e^{-2y}\cdot e^{-1}\,dy = \frac{2}{e}\int_{0}^{1} e^{-2y}\,dy$$

$$= \frac{2}{e}\left[\frac{-e^{-2y}}{2}\right]_{y=0}^{1}$$

$$= \frac{e^0 - e^{-2}}{e}$$

$$= \frac{1}{e} - \frac{1}{e^3}$$

$\int e^{-2x}\,dx$

$\frac{1}{2}\int e^{-t}\,dt \quad 2x=t \Rightarrow dx=\frac{dt}{2}$

$-\frac{e^{-t}}{2}$

$-\frac{e^{-2x}}{2}$

b. Calculate P(X<Y).

$$P\{X<Y\} = \iint_{x,y:\,x<y} f(x,y)\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\left[\int_{-\infty}^{y} f(x,y)\,dx\right]dy$$

$$= \int_{0}^{\infty}\left[\int_{0}^{y} 2e^{-x}e^{-2y}\,dx\right]dy$$

$$= \int_{0}^{\infty} 2e^{-2y}\left[-e^{-x}\right]_{x=0}^{y} dy$$

$$= 2\int_{0}^{\infty} e^{-2y}\cdot(1-e^{-y})\,dy$$

$$= 2\int_{0}^{\infty}(e^{-2y} - e^{-3y})\,dy$$

$$= 2\left[\frac{e^{-2y}}{-2} - \frac{e^{-3y}}{-3}\right]_{0}^{\infty}$$

$$= 2\left[\frac{-1}{-2} + \frac{1}{-3}\right] = 2\left[\frac{1}{2} - \frac{1}{3}\right] = 2\cdot\frac{1}{6} = \frac{1}{3}$$

Marginal probability density functions

Given the joint PDF of 2 rvs f(X, Y), we can calculate the marginal density functions as

$$f_x(x) = \int_{-\infty}^{\infty} f(x,y)\,dy \qquad \text{By, } f_y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

c. Calculate P(X<a).

$$P\{X<a\} = \int_0^{\infty}\int_0^{a} f(x,y)\,dx\,dy$$

OR using marginal probability

$$P_x(x) = \int_{-\infty}^{\infty} f(x,y)\,dy \qquad \therefore \text{ Marginal PDF : } P_x(x) = \begin{cases} e^{-x}, & x>0 \\ 0, & \text{otherwise} \end{cases}$$

$$= \int_0^{\infty} 2e^{-x}e^{-2y}\,dy$$

$$= 2e^{-2x}\left[\frac{e^{-2y}}{-2}\right]_{y=0}^{\infty}$$

$$= e^{-x}$$

$$P(X<a) = \int_{-\infty}^{a} f(x)\,dx$$

$$= \int_0^{a} e^{-x}\,dx$$

$$= 1-e^{-a}$$

Independent Random Variables

In the jointly continuous case, the condition of independence is equivalent to

$$f(x,y) = f_x(x) \cdot f_y(y)$$

Q. If the joint probability density of X and Y is

$$f(x,y) = \begin{cases} 6e^{-2x}e^{-3y}, & 0<x<\infty, 0<y<\infty \\ 0, & \text{otherwise} \end{cases}$$

Are X and Y independent?

→ $f_x(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$

$= \int_0^{\infty} 6e^{-2x}e^{-3y}\,dy = 6e^{-2x}\left[\frac{e^{-3y}}{-3}\right]_0^{\infty}$

$= -2e^{-2x}[0-1] = 2e^{-2x}$

$$f_x(x) = \begin{cases} 2e^{-2x}, & 0<x<\infty \\ 0, & \text{else} \end{cases}$$

Now $f_y(y) = \int_0^{\infty} 6e^{-2x}e^{-3y}\,dx = 3e^{-3y}$

$$f_y(y) = \begin{cases} 3e^{-3y}, & 0<y<\infty \\ 0, & \text{else} \end{cases}$$

$\therefore f(x,y) = f_x(x) \cdot f_y(y)$

$\therefore$ X and Y are independent.

Conditional distributions

If X and Y have a joint PDF f(X, Y), then the conditional probability density function of X given Y=y is defined for all values of y such that

$$f_{x|y}(x|y) = \frac{f(x,y)}{f_y(y)}$$

For any set A,

$$P\{X \in A\} = \int_A f(x)\,dx$$

$$P\{X \in A \mid Y=y\} = \int_A \frac{f(x,y)}{f_y(y)}\,dx$$

Now, $F_{x|y}(a|y) = \int_{-100}^{a} \frac{f(x,y)}{f_y(y)}\,dx$ ← conditional cumulative density function.

Q. The joint density function of X and Y is given as

$$f(x,y) = \begin{cases} \frac{12}{5}x(2-x-y), & 0<x<1, 0<y<1 \\ 0, & \text{otherwise} \end{cases}$$

Compute the conditional density of X, given that Y=k, 0<k<1.

→ $f_{x|y}(x|y=k) = \frac{f(x,y)}{f_y(k)}$

↳ $f_y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_0^1 \frac{12}{5}x(2-x-y)\,dx = \frac{12}{5}\int_0^1 (2x - x^2 - xy)\,dx$

$= \frac{12}{5}\left[x^2 - \frac{x^3}{3} - y\frac{x^2}{2}\right]_{x=0}^{1} = \frac{12}{5}\left[1 - \frac{1}{3} - \frac{y}{2}\right]$

$= \frac{12}{5}\left[\frac{2}{3} - \frac{y}{2}\right] = \frac{12}{5}\left[\frac{4-3y}{6}\right]$

$\therefore f_y(k) = \frac{12}{5}\left(\frac{4-3k}{6}\right)$

$f_{x|y}(x|k) = \frac{\frac{12}{5}x(2-x-k)}{\frac{12}{5}\left(\frac{4-3k}{6}\right)} = \frac{6x(2-x-k)}{4-3k}$

$\therefore f_{x|y}(x|k) = \begin{cases} \frac{6x(2-x-k)}{(4-3k)}, & 0<x<1 \\ 0, & \text{otherwise} \end{cases}$

Q. Suppose that the joint density of X and Y is given as

$$f(x,y) = \begin{cases} \dfrac{e^{-x/y}\, e^{-y}}{y}, & 0 < x < \infty,\ 0 < y < \infty \\ 0, & \text{otherwise} \end{cases}$$

Find $P\{X > 1 \mid Y = 2\}$

→ $f_{X|Y}(x|y) = \dfrac{f(x,y)}{f_Y(y)}$

$$f_{X|Y}(x|y) = \frac{e^{-x/y}\, e^{-y}}{y} \times \frac{1}{e^{-y}}$$

$$= \frac{e^{-x/y}}{y}$$

$$f_{X|Y}(x|y) = \begin{cases} \dfrac{e^{-x/y}}{y}, & 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_0^{\infty} \frac{e^{-x/y}\, e^{-y}}{y}\,dx$$

$$= \frac{e^{-y}}{y} \int_0^{\infty} e^{-x/y}\,dx = \frac{e^{-y}}{y}\left[\frac{e^{-x/y}}{-1/y}\right]_{x=0}^{\infty}$$

$$= -e^{-y}\left[0 - 1\right] = e^{-y}$$

$$f_Y(y) = \begin{cases} e^{-y}, & 0 < y < \infty \\ 0, & \text{otherwise} \end{cases}$$

$$P(X > 1 \mid Y = 2) = \int_1^{\infty} \frac{e^{-x/2}}{2}\,dx = \frac{1}{2}\left[\frac{e^{-x/2}}{-1/2}\right]_{x=1}^{\infty} = -\left[0 - e^{-1/2}\right] = e^{-1/2}$$

If X and Y are independent continuous random variables,
the conditional density of X given Y=y is just the unconditional density of X.

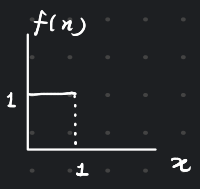
$$f_{X|Y}(x|y) = f_X(x)$$

Proof: $f_{X|Y}(x|y) = \dfrac{f(x,y)}{f_Y(y)} = \dfrac{f_X(x) \cdot f_Y(y)}{f_Y} = f_X(x)$ → as X and Y are independent.

Distribution of function of a random variable

Q. The density function of X is given by

$$f(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$



Calculate the density function of $Y = e^X$.

→

Start by calculating the cumulative density function of Y, $F_Y(y)$.

$$F_Y(y) = P\{Y \le y\}$$
$$= P\{e^X \le y\}$$
$$= P\{X \le \log y\} = F_X(\log y)$$
$$= \int_0^{\log y} f_X(x)\,dx$$

$$f_X(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

when $0 \le x \le 1$,
$1 \le y \le e$ as $y = e^x$

$$F_Y(y) = \int_0^{\log y} 1\,dx$$
$$= [x]_0^{\log y}$$
$$= \log y$$

otherwise, when $y > e$, $F_Y(y) = 1$
when $y < 1$, $F_Y(y) = 0$

$$F_Y(y) = \begin{cases} \log y, & 1 \le y \le e \\ 0, & y < 1 \\ 1, & y > e \end{cases}$$

differentiate →

$$f_Y(y) = \begin{cases} 1/y, & 1 \le y \le e \\ 0, & \text{otherwise} \end{cases}$$

Q. A r.v. X has the density func^n: $f_x(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{else} \end{cases}$
calc. that for $Y = X^2$.

→ $F_y(y) = P\{Y \le y\}$
$= P\{X^2 \le y\}$
$= P\{X \le \sqrt{y}\}$
$= F_x(\sqrt{y})$
$= \int_0^{\sqrt{y}} f_x(x)$

For $0 \le y \le 1$, $F_y(y) = \int_0^{\sqrt{y}} 1 \, dx = [x]_0^{\sqrt{y}} = \sqrt{y}$

For $y < 0$, $F_y(y) = 0$
$y > 1$, $F_y(y) = 1$

$$F_y(y) = \begin{cases} \sqrt{y}, & 0 \le y \le 1 \\ 0, & y < 0 \\ 1, & y > 1 \end{cases}$$

↳ $$f_y(y) = \begin{cases} \frac{1}{2} y^{-1/2}, & 0 \le y \le 1 \\ 0, & \text{else} \end{cases}$$

$$f_y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & 0 \le y \le 1 \\ 0, & \text{else} \end{cases}$$

Q. Same ques^n for $Y = X^n$

$F_y(y) = P\{Y \le y\}$
$= P\{X^n \le y\}$
$= P\{X \le y^{1/n}\}$
$= F_x(y^{1/n})$
$= \int_{-\infty}^{y^{1/n}} f(x) \, dx$

for $0 \le y \le 1$, $F_y(y) = \int_{-\infty}^{y^{1/n}} 1 \, dx = y^{1/n}$

$f_y(y) = \frac{1}{n} y^{\frac{1}{n}-1}$

$$f_y(y) = \begin{cases} \frac{1}{n} y^{\frac{1}{n}-1}, & 0 \le y \le 1 \\ 0, & \text{else} \end{cases}$$

eg: $f_X(x) = \begin{cases} e^x, & 0 \le x \le 1 \\ Ax, & 2 \le x \le 3 \\ 0, & \text{otherwise} \end{cases}$

(i) Find the value of A

(ii) Probability density funct
$Y = X^2$

Expectation of continuous random variables

We defined the expected value of a discrete rv X by $E[X] = \sum_x x \cdot p(x=x)$.

For a continuous rv with pdf f(x),

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

Q. The pdf of a continuous rv X is $f(x) = \begin{cases} 2x, & \text{if } 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$
Find E[X].

→ $E[X] = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$

$= \int_0^1 x \cdot 2x + 0 + 0$

$= 2\left[\frac{x^3}{3}\right]_{x=0}^{1} = \frac{2}{3}$.

Q. The pdf of a continuous rv Z is $f_Z(z) = 2(1-z)$ if $0 \le z \le 1$.
Find E[Z].

→ $E[Z] = \int_0^1 z \cdot 2(1-z)\, dz = 2\left[\frac{z^2}{2} - \frac{z^3}{3}\right]_0^1 = 2\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{1}{3}$

Q. Suppose you're expecting a message some time after 5:00 pm. X, the # of hours after 5 pm until the message arrives, is a random variable.

$f_x(x) = \begin{cases} 1/1.5 & 0 \le x \le 1.5 \\ 0 & \text{otherwise} \end{cases}$

What is the expected time when the message arrives?

→ $E[X] = \int_0^{1.5} x \cdot \frac{1}{1.5}\, dx = \frac{1}{1.5}\left[\frac{x^2}{2}\right]_0^{1.5} = \frac{1}{1.5} \times \frac{(1.5)^2}{2} = \frac{1.5}{2} = 0.75 \text{ hrs} = \frac{3}{4} \text{ hrs} = 45 \text{ mins}$.

∴ time at which the message arrives is 5:45 p.m.

Q. The density function of a c.r.v. is given as $f_x(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$
Find E[exp(X)].

→ $Y = e^X$. To find $f_Y(y)$, we first find $F_Y(y)$.

$F_Y(y) = P\{Y \le y\}$

$= P\{e^X \le y\}$

$= P\{X \le \ln y\} = P_x(\log y)$

$= \int_{-\infty}^{\ln y} f(x)\, dx$ $\qquad 0 \le x \le 1 \rightarrow 1 \le y \le e$

$= \int_0^{\log y} 1 \cdot dx = \log y$

∴ $f_Y(y) = \frac{d}{dy} \log y = \frac{1}{y}$ $\qquad F_Y(y) = \begin{cases} 1/y & 1 \le y \le e \\ 0 & \text{otherwise} \end{cases}$

$E[Y] = \int_1^e y \cdot \frac{1}{y}\, dy = e - 1$

Q. The time taken, in hours, to repair a breakdown in a factory is a rv of which the density function is

$$f_x(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

If the cost involved in a breakdown for x hours is x^3 dollars, what is the expected cost of a breakdown?

→ $y = x^3$

$$F_y(y) = P\{x^3 \le y\}$$
$$= P\{x \le y^{1/3}\}$$

$$\therefore f_y(y) = \begin{cases} \frac{1}{3} y^{-2/3}, & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$$f_y(y) = \begin{cases} \frac{1}{n} y^{\frac{1}{n}-1}, & 0 \le y \le 1 \\ 0, & \text{else} \end{cases}$$

$$E[y] = \int_0^1 y \cdot \frac{1}{3} \cdot y^{-2/3} \, dy = \frac{1}{3} \int_0^1 y^{1/3} \, dy = \frac{1}{3} \left[\frac{y^{4/3}}{4/3} \right]_0^1 = \frac{1}{4}.$$

Law of the unconscious statistician (LOTUS)

The form of the law depends on the type of random variable X in question. If the distribution of X is discrete and one knows its probability mass function p_X, then the expected value of $g(X)$ is

$$\mathrm{E}[g(X)] = \sum_x g(x) p_X(x),$$

where the sum is over all possible values x of X. If instead the distribution of X is continuous with probability density function f_X, then the expected value of $g(X)$ is

$$\mathrm{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,\mathrm{d}x$$

same thing using LOTUS

Q. The time taken, in hours, to repair a breakdown in a factory is a rv of which the density function is

$$f_x(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

If the cost involved in a breakdown for x hours is x^3 dollars, what is the expected cost of a breakdown?

→ $y = x^3$

$$E[y] = \int_{-\infty}^{\infty} g(x) \cdot f_x(x)\,dx$$

$$= \int_0^1 x^3 \cdot 1 \cdot dx = \left[\frac{x^4}{4}\right]_0^1 = \frac{1}{4}$$

Linearity of Expectations

$$E[aX+b] = aE[X] + b$$

Proof: $E[aX+b] = \int_{-\infty}^{\infty} (ax+b) f(x)\,dx$

$= a\int_{-\infty}^{\infty} x \cdot f(x)\,dx + b\int_{-\infty}^{\infty} f(x)\,dx$

$= aE[X] + b$

For joint distributions

For 2 rvs X and Y whose joint PDF is given by f(x, y), then for any real-valued function g,

$$E[g(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y) \cdot f(x,y)\,dx\,dy$$

Linearity of Expectations

$$E[X+Y] = E[X] + E[Y]$$

Proof:

$E[X+Y] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x+y) f(x,y)\,dx\,dy$

$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x \cdot f(x,y)\,dy\,dx + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} y \cdot f(x,y)\,dx\,dy$

$= \int_{-\infty}^{\infty} x\left[\int_{-\infty}^{\infty} f(x,y)\,dy\right]dx + \int_{-\infty}^{\infty} y\left[\int_{-\infty}^{\infty} f(x,y)\,dx\right]dy$

$= \int_{-\infty}^{\infty} x \cdot f_X(x)\,dx + \int_{-\infty}^{\infty} y \cdot f_Y(y)\,dy$

$= E[X] + E[Y]$.

In general for any integral value $n \geq 1$,

$$E[X_1 + \cdots + X_n] = E[X_1] + \cdots + E[X_n]$$

Let X and Y be independent real-valued CRVs with finite expected values. Then,

$$E[XY] = E[X] \cdot E[Y]$$

Proof:

$E[XY] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f(x,y)\,dx\,dy$

$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy \cdot f(x) f(y)\,dx\,dy$ .. as X & Y are independent.

$= \int_{-\infty}^{\infty} x \cdot f(x)\,dx \cdot \int_{-\infty}^{\infty} y \cdot f(y)\,dy$

$= E[X] \cdot E[Y]$

In general, for n mutually independent rvs X1, ..., Xn,

$$E[X_1 \cdots X_n] = E[X_1] \cdots E[X_n]$$

Q. Let X and Y be two jointly continuous rvs with joint PDF as

$$f(x,y) = \begin{cases} x+y, & 0 \le x, y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find E[XY^2].

→ $E[g(x,y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y) \cdot f(x,y)\, dx\, dy$

$$= \int_0^1 \int_0^1 xy^2 (x+y)\, dx\, dy$$

$$= \int_0^1 \left[x^2 \frac{y^3}{3} + x \frac{y^4}{4} \right]_{y=0}^{1} dx$$

$$= \int_0^1 \left(\frac{x^2}{3} + \frac{x}{4} \right) dx$$

$$= \left[\frac{x^3}{9} + \frac{x^2}{8} \right]_{x=0}^{1}$$

$$= \frac{17}{72}$$

Conditional expectation

If X and Y are continuous rvs, then X | Y=y is also a continuous rv with density f(x|y)

$$f_{x|y}(x|y) = \frac{f(x,y)}{f_y(y)}$$

The conditional expectation of X | Y=y is

$$E[X] = \int_{-\infty}^{\infty} x \cdot f_{x|y}(x|y)\,dx$$

$$= \int_{-\infty}^{\infty} x \cdot \frac{f(x,y)}{f_y(y)}\,dx, \qquad f_y(y) \neq 0$$

Q. The joint density function of X and Y is

$$f(x,y) = \frac{e^{-x/y}\, e^{-y}}{y}, \qquad 0 < x, y < \infty$$

Compute E[X|Y=y].

→ $$f_y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_0^{\infty} \frac{e^{-x/y}\, e^{-y}}{y}\,dx$$

$$= \frac{e^{-y}}{y}\left[\frac{e^{-x/y}}{-1/y}\right]_{x=0}^{\infty}$$

$$= e^{-y}$$

$$f_{x|y}(x|y) = \frac{f(x,y)}{f_y(y)} = \frac{e^{-x/y}\, e^{-y}}{y \cdot e^{-y}} = \frac{e^{-x/y}}{y}, \qquad 0 < x < \infty$$

$$f_{x|y}(x|y) = \begin{cases} \frac{e^{-x/y}}{y}, & \text{if } 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

$$E[X|Y=y] = \int_{-\infty}^{\infty} x \cdot f_{x|y}(x|y)\,dx$$

$$= \int_0^{\infty} x \cdot \frac{e^{-x/y}}{y}\,dx$$

$$= \frac{1}{y}\int_0^{\infty} x \cdot e^{-x/y}\,dx$$

$$= \frac{1}{y}\left[\frac{e^{-x/y}}{-1/y}\cdot[x+y]\right]_{x=0}^{\infty}$$

$$= \left[e^{-x/y}(x+y)\right]_{x=\infty}^{0}$$

$$= y$$

$\int x e^{ax}\,dx$ (I: x, II: e^{ax}) — ILATE

$u\int v\,dx - \int u' \int v\,dx$

$= x\frac{e^{ax}}{a} - \int 1 \cdot \frac{e^{ax}}{a}\,dx$

$= \frac{x}{a}e^{ax} - \frac{1}{a}\cdot\frac{e^{ax}}{a}$

$= \frac{e^{ax}}{a}\left[x - \frac{1}{a}\right]$

LOTUS

- $E[g(X) \mid Y=y] = \begin{cases} \sum_x g(x) \cdot P_{X|Y}(x|y) & \text{if } X, Y \text{ are DRVs} \\ \int_{-\infty}^{\infty} g(x) \cdot f_{X|Y}(x|y) & \text{if } X, Y \text{ are CRVs} \end{cases}$
- $E[X_1 + X_2 \mid Y=y] = E[X_1 \mid Y=y] + E[X_2 \mid Y=y]$

Q. Let X, Y be 2 jointly CRVs with PDF $f_{X,Y}(x,y) = \begin{cases} \frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{6}, & 0 \le x \le 1, \ 0 \le y \le 2 \\ 0, & \text{otherwise} \end{cases}$

a) For $0 \le k \le 2$, find conditional PDF of X given $Y = k$.

b) $P(X < \frac{1}{2} \mid Y = 1)$

c) $E[X \mid Y = 1]$

d) $E[X^2 \mid Y = 1]$

→ $f_{XY}(x,y) = \begin{cases} \frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{6}, & 0 \le x \le 1, \ 0 \le y \le 2 \\ 0 & \text{otherwise} \end{cases}$

(a) $f_{X|Y}(x|y) = \frac{f(x,k)}{f_Y(k)}$ (k)

$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$

$= \int_0^1 (\frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{6})\,dx$

$= \left[\frac{x^3}{12} + \frac{xy^2}{4} + \frac{x^2 y}{12}\right]_{x=0}^{1}$

$= \frac{1}{12} + \frac{y^2}{4} + \frac{y}{12}$

$= \frac{1}{12}[1 + 3y^2 + y] \qquad 0 \le y \le 2$

$f_{X|Y}(x|k) = \begin{cases} \frac{3x^2 + 3k^2 + 2xk}{3k^2 + k + 1}, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$

b) $P(X < \frac{1}{2} \mid Y = 1)$

→ $f_{X|Y}(x|Y=1) = \begin{cases} \frac{3x^2 + 2x + 3}{5} & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$

$P(X < \frac{1}{2} \mid Y = 1) = \int_{-\infty}^{1/2} f_{X|Y}(x|Y=1)$

$= \int_0^{1/2} \frac{3x^2 + 2x + 3}{5}\,dx$

$= \frac{1}{5}\left[x^3 + x^2 + 3x\right]_0^{1/2}$

$= \frac{1}{5}\left[\frac{1}{8} + \frac{1}{4} + \frac{3}{2}\right]$

$= \frac{1}{40}[1 + 2 + 12] = \frac{15}{40} = \frac{3}{8}$.

c) $E[X \mid Y = 1] = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y=1)\,dx$

$= \int_0^1 x\,(\frac{3x^2 + 2x + 3}{5})\,dx = \frac{1}{5}\left[\frac{3x^4}{4} + \frac{2x^3}{3} + \frac{3x^2}{2}\right]_0^1$

$= \frac{1}{5}\left[\frac{9 + 8 + 18}{12}\right] = \frac{7}{12}$.

d) $E[X^2 \mid Y = 1]$

$= \int_{-\infty}^{\infty} x^2 \cdot f_{X|Y}(x|Y=1)\,dx$

$= \int_0^1 x^2 \cdot \frac{3x^2 + 2x + 3}{5}\,dx = \frac{1}{5}\left[\frac{3x^5}{5} + 2\frac{x^4}{4} + 3\frac{x^3}{3}\right]_0^1$

$= \frac{1}{5}\left[\frac{3}{5} + \frac{1}{2} + 1\right] = \frac{1}{5} \cdot 10\,(6 + 5 + 10) = \frac{21}{50}$

Law of Total Probability

$$P(A) = \int_{-\infty}^{\infty} P(A \mid X = x) \cdot f_X(x)\, dx$$

Q. Let X and Y be 2 independent CRVs with PDF $f_X(x) = \begin{cases} 1 & 0 \le x \le 1 \\ 0 & \text{else} \end{cases}$, $f_Y(y) = \begin{cases} 1 & 0 \le y \le 1 \\ 0 & \text{else} \end{cases}$.
Find $P(X^3 + Y \ge 1)$

→ As X and Y are independent,

$$f_{XY}(x,y) = f_X(x) \cdot f_Y(y) = 1 \cdot 1 = 1$$

$$f_{XY}(x,y) = \begin{cases} 1 & 0 \le x, y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$P(X^3 + Y > 1) = P(Y > 1 - X^3)$$

$$= \int_0^1 \int_{1-x^3}^1 1 \cdot dy\, dx = \int_0^1 [y]_{1-x^3}^1 dx = \int_0^1 (1 - 1 + x^3)\, dx$$

$$= \left[\frac{x^4}{4}\right]_{x=0}^1 = \frac{1}{4}$$

OR using total probability,

$$P(A) = \int_{-\infty}^{\infty} P(A \mid X = x) \cdot f_X(x) \cdot dx$$

$$P(X^3 + Y > 1) = \int_{-\infty}^{\infty} P(X^3 + Y > 1 \mid X = x) \cdot f_X(x)\, dx$$

$$= \int_0^1 P(x^3 + Y > 1 \mid X = x) \cdot 1\, dx$$

$$= \int_0^1 P(Y > 1 - x^2 \mid X = x)\, dx$$

$$= \int_0^1 P(Y > 1 - x^3)\, dx \qquad \text{since } X, Y \text{ are independent.}$$

$$= \int_0^1 \int_{1-x^3}^1 1 \cdot dy\, dx = \frac{1}{4}$$

Law of Iterated Expectations

$$E[E[X|Y]] = E[X]$$

Proof: $f_{X|Y}(x|Y=y) = \frac{f(x,y)}{f_Y(y)}$

$$E[X|Y=y] = \int_{-\infty}^{\infty} x \cdot \frac{f(x,y)}{f_Y(y)}\,dx = g(y) \quad \text{some func}^n \text{ of } y.$$

$$E[E[X|Y]] = \int_{-\infty}^{\infty} E[X|Y=y] \cdot f_Y(y)\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x \cdot \frac{f(x,y)}{f_Y(y)}\,dx\, f_Y(y)\,dy$$

$$= \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f(x,y)\,dy\right] dx$$

$$= \int_{-\infty}^{\infty} x\, f_X(x)\,dx$$

$$= E[X]$$

Q. $f(x,y) = x+y$ for $0 \le x \le 1$, $0 \le y \le 1$. Find $E[E[x|y]]$. Verify if it is equal to $E[x]$.

→ $f_x(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$

$= \int_0^1 (x+y)\,dy = \left[xy + \frac{y^2}{2}\right]_0^1 = x + \frac{1}{2}$

$$f_x(x) = \begin{cases} x+1/2 & 0 \le x \le 1 \\ 0 & \text{else} \end{cases}$$

Ily, $f_y(y) = \begin{cases} y+1/2 & 0 \le y \le 1 \\ 0 \end{cases}$

$f_{x|y}(x|y) = \frac{f(x,y)}{f(y)}$

$= \frac{x+y}{y+1/2}$ for $0 \le y \le 1$, $0 \le x \le 1$

$$f_{x|y}(x|y) = \begin{cases} \frac{2(x+y)}{1+2y} & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & \text{else} \end{cases}$$

$E[x|y=y] = \int_{-\infty}^{\infty} x \cdot f_{x|y}(x|y)\,dx$

$= \int_0^1 x \cdot \frac{2(x+y)}{1+2y}\,dx$

$= \frac{2}{1+2y} \int_0^1 x \cdot (x+y)\,dx$

$= \frac{2}{1+2y}\left[\frac{x^3}{3} + \frac{x^2 y}{2}\right]_0^1 = \frac{2}{1+2y}\,\frac{(2+3y)}{6}$

$E[x|y=y] = \frac{2+3y}{3+6y}$

$E[E[x|y=y]] = \int_{-\infty}^{\infty} E[x|y=y] \cdot f_y(y)\,dy$

$= \int_0^1 \frac{2+3y}{3+6y}\,(1/2+y)\,dy$

$= \int_0^1 \frac{2+3y}{3(1+2y)}\,\frac{(1+2y)}{2}\,dy$

$= \frac{1}{3\cdot 2}\int_0^1 (2+3y)\,dy$

$= \frac{1}{6}\left[2y + \frac{3y^2}{2}\right]_0^1 = 7/12$

$E[x] = \int_0^1 (1/2+x)\,x\,dx$

$= \left[\frac{x^2}{4} + \frac{x^3}{3}\right]_0^1 = 7/12$

Both are equal.

Variance of CRVs

$\text{Var}[X] = E[X-\mu]^2 = \int_{-\infty}^{\infty} (x-\mu)^2 \cdot f_X(x)\,dx$

$= E[X^2] - E[X]^2 = \int_{-\infty}^{\infty} x^2 \cdot f_X(x)\,dx - \left(\int_{-\infty}^{\infty} x \cdot f_X(x)\,dx\right)^2$

Q. Find $\text{var}(X)$ when the density of X is given by $f(x) = \begin{cases} 2x, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$.

→ $\text{var}(X) = \int_{-\infty}^{\infty} x^2 \cdot f_X(x)\,dx - \left(\int_{-\infty}^{\infty} x \cdot f_X(x)\,dx\right)^2$

$= \int_0^1 x^2 \cdot (2x)\,dx - \left(\int_0^1 x \cdot (2x)\,dx\right)^2$

$= 2\left[\frac{x^4}{4}\right]_0^1 - 4\left(\left[\frac{x^3}{3}\right]_0^1\right)^2 = 2 \cdot \frac{1}{4} - 4 \cdot \frac{1}{9} = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$

Standard Deviation $\sigma = \sqrt{\text{var}(X)}$

$\text{Var}(aX+b) = a^2 \text{var}(X)$

If X and Y are independent, $\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$

Conditional Variance

$\text{Var}(X|Y=y) = E\left[(X - E[X|Y=y])^2 \,|\, Y=y\right]$

$= E[X^2|Y=y] - E[X|Y=y]^2$

$= \int_{-\infty}^{\infty} x^2 f_{X|Y}(x|y)\,dx - \left[\int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y)\,dx\right]^2$

Q. Suppose the CRVs X and Y have joint PDF: $f(x,y) = \frac{3}{2}$, $x^2 \le y \le 1$, $0 \le x \le 1$.
Find conditional var of Y given $X=x$.

→ $\text{Var}(Y|X=x) = E[Y^2|X=x] - E[Y|X=x]^2$

$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_{x^2}^{1} \frac{3}{2}\,dy = \frac{3}{2}(1-x^2)$, $0 \le x \le 1$ $\quad = 0$ otherwise

$f_{Y|X}(y|X=x) = \frac{f(x,y)}{f_X(x)} = \frac{\frac{3}{2}}{\frac{3}{2}(1-x^2)} = \frac{1}{(1-x^2)}$ if $0 \le x \le 1$, $1 \le y \le x^2$

0 otherwise

e.g. $f_{Y|X}(Y|X=\tfrac{1}{2}) = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}$, $\frac{1}{4} \le y \le x^2$

0, otherwise.

$E[Y|X=x] = \int_{-\infty}^{\infty} y \cdot f_{Y|X}(y|x)\,dy = \int_{x^2}^{1} \frac{1}{1-x^2} y\,dy = \frac{1}{1-x^2}\left[\frac{y^2}{2}\right]_{x^2}^1 = \frac{1}{1-x^2}\cdot\frac{(1-x^4)}{2} = \frac{1+x^2}{2}$

e.g. $E[Y|X=\tfrac{1}{2}] = \frac{1+\frac{1}{4}}{2} = \frac{5}{8}$

$E[Y^2|X=x] = \int_{-\infty}^{\infty} y^2 \cdot f_{Y|X}(y|x)\,dy = \int_{x^2}^{1} y^2 \cdot \frac{1}{1-x^2}\,dy = \frac{1}{1-x^2}\left[\frac{y^3}{3}\right]_{x^2}^1 = \frac{1}{1-x^2}\cdot\frac{1}{3}\cdot(1-x^6) = \frac{(1-x^2)(1+x^2+x^4)}{3(1-x^2)} = \frac{1+x^2+x^4}{3}$

$\text{Var}(Y|X=x) = \frac{1+x^2+x^4}{3} - \left(\frac{1+x^2}{2}\right)^2 = \ldots = \frac{(x^2-1)^2}{12}$

Law of total variance

$$\text{Var}(X) = \text{Var}(E[X|Y]) + E[\text{var}(X|Y)]$$

Q. $E[X|Y=y] = 3y$; $\text{var}(X|Y=y) = 2$; $\text{var}(Y) = 1/9$. Find $\text{var}(X)$.

→ $\text{var}(X) = \text{Var}(E[X|Y]) + E[\text{var}(X|Y)]$

$= \text{var}(3y) + E[2]$

$= 9\,\text{var}(Y) + 2$

$= 9 \cdot \frac{1}{9} \cdot 2 = 3$

Covariance of a continuous rv

$$\text{cov}(X, Y) = E[(X-\mu_X)(Y-\mu_Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x-\mu_x)(y-\mu_y) f_{xy}(x,y)\,dx\,dy$$

$$= E[XY] - E[X]\cdot E[Y] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy \cdot f_{x,y}(x,y)\,dx\,dy - \int_{-\infty}^{\infty} x\, f_x(x)\,dx \cdot \int_{-\infty}^{\infty} y \cdot f_y(y)\,dy$$

Q. Joint PDF $f_{Y_1,Y_2}(y_1,y_2) = \begin{cases} \frac{3}{16} y_1^2, & 0 \le y_1, y_2 \le 2 \\ 0, & \text{otherwise} \end{cases}$. compute $\text{cov}(Y_1, Y_2)$.

→ $f_{Y_1}(y_1) = \int_0^2 \frac{3}{16} y_1^2\,dy_2 = \frac{3}{16} y_1^2 [y_2]_0^2 = \frac{3}{8} y_1^2$.

$f_{Y_2}(y_2) = \int_0^2 \frac{3}{16} y_1^2\,dy_1 = \frac{3}{16}\left[\frac{y_1^3}{3}\right]_0^2 = \frac{1}{16} \cdot 8 = \frac{1}{2}$

$\text{cov}(X, Y) = E[XY] - E[X]\cdot E[Y]$

$\text{cov}(Y_1, Y_2) = \int_0^2\int_0^2 y_1 \cdot y_2 \cdot \frac{3}{16} y_1^2\,dy_1\,dy_2 - \int_0^2 y_1 f_{Y_1}(y_1)\,dy_1 \int_0^2 y_2 f_{Y_2}(y_2)\,dy_2$

$= \int_0^2 \frac{3}{16} y_2 \cdot \left[\frac{y_1^4}{4}\right]_0^2 dy_2 - \int_0^2 y_1 \cdot \frac{3}{8} y_1^2\,dy_1 \int_0^2 y_2 \cdot \frac{1}{2}\,dy_2$

$= \frac{3}{16}\int_0^2 y_2 \cdot \frac{16}{4}\,dy_2 - \frac{3}{8}\left[\frac{y_1^4}{4}\right]_0^2 \cdot \frac{1}{2}\left[\frac{y_2^2}{2}\right]_0^2$

$= \frac{3}{4}\int_0^2 y_2\,dy_2 - \frac{3}{8}\cdot\frac{16}{4} \cdot \frac{1}{2}\cdot\frac{4}{2}$

$= \frac{3}{4}\left[\frac{y_2^2}{2}\right]_0^2 - \frac{3}{2}$

$= \frac{3}{4}\cdot\frac{4}{2} - \frac{3}{2}$

$= 0$

$\text{cov}(Y_1, Y_2) = 0$

$\therefore$ Y_1 and Y_2 are independent.

Propositions:

i. $\text{cov}(X,Y) = \text{cov}(Y,X)$

ii. $\text{cov}(X,X) = \text{var}(X)$

Proof: $\text{cov}(X,X) = E[(x-\mu_X)(x-\mu_X)]$
$= E[(x-\mu_X)^2]$
$= \text{var}(X)$

iii. $\text{cov}(aX,Y) = a \cdot \text{cov}(X,Y)$

Proof: $E[(ax-a\mu_X)(y-\mu_Y)]$
$= a\,E[(x-\mu_X)(y-\mu_Y)]$

iv. $\text{cov}(X+Y,Z) = \text{cov}(X,Z) + \text{cov}(Y,Z)$

Proof: $\text{cov}(X+Y,Z)$
$= E[(X+Y)Z] - E[X+Y]E[Z]$
$= E[XZ+YZ] - (E[X]+E[Y])E[Z]$
$= E[XZ] - E[X]E[Z] + E[YZ] - E[Y]E[Z]$
$= \text{cov}(X,Z) + \text{cov}(Y,Z)$

v. generalization of the previous:

$$\text{Cov}\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{m} \text{cov}(X_i, Y_j)$$

i.e. covariance is additive.

$\mu_i \to$ mean of X_i
$\nu_i \to$ mean of Y_i

Proof: $\text{cov}(\Sigma X_i, \Sigma Y_i) = E[(\Sigma X_i - E[\Sigma X_i])(\Sigma Y_j - E[\Sigma Y_j])]$
$= E[(\Sigma X_i - \Sigma \mu_i)(\Sigma Y_j - \Sigma \nu_j)]$
$= E[\Sigma(X_i - \mu_i) \cdot \Sigma(Y_j - \nu_j)]$
$= E[\sum_i \sum_j (X_i - \mu_i)(Y_j - \nu_j)]$
$= \Sigma\Sigma\, E[(X_i - \mu_i)(Y_j - \nu_j)]$
$= \Sigma\Sigma\, \text{cov}(X_i, Y_i)$

vi. $\text{cov}(X+c, Y) = \text{cov}(X,Y)$, c is a constant

Proof: $\text{cov}(X+c, Y)$
$= E[(X+c - \mu_{X+c})(Y-\mu_Y)]$
$= E[(X+c-(\mu_X+c))(Y-\mu_Y)]$
$= E[(X-\mu_X)(Y-\mu_Y)] = \text{cov}(X,Y)$

vii. $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{cov}(X,Y)$

Proof: $\text{var}(X+Y) = \text{cov}(X+Y, X+Y)$
$= \text{cov}(X,X) + \text{cov}(X,Y) + \text{cov}(Y,X) + \text{cov}(Y,Y)$
$= \text{var}(X) + 2\,\text{cov}(X,Y) + \text{var}(Y)$

same for discrete random variables

viii. Generalization of the previous one:

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{var}(X_i) + 2\sum_{i<j}\sum \operatorname{cov}(X_i, X_j)$$

proof: $\operatorname{var}\left(\sum_{i=1}^{n} X_i\right)$

$= \operatorname{cov}\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{n} X_j\right)$

$= \sum_{i=1}^{n}\sum_{j=1}^{n} \operatorname{cov}(X_i, X_j)$

$= \operatorname{cov}(X_1, X_1) + \operatorname{cov}(X_1, X_2) + \cdots + \operatorname{cov}(X_1, X_n)$

$\operatorname{cov}(X_2, X_1) + \operatorname{cov}(X_2, X_2) + \cdots + \operatorname{cov}(X_2, X_n)$

$+$

$\vdots$

$+$

$\operatorname{cov}(X_n, X_1) + \cdots + \operatorname{cov}(X_n, X_n)$ $\leftarrow \sum \operatorname{var}(X_i)$

$= \sum \operatorname{var}(X_i) + 2\sum_{i<j}\sum \operatorname{cov}(X_i, X_j)$

ix. $\operatorname{Var}(aX + bY + c) = a^2 \operatorname{var}(X) + b^2 \operatorname{var}(Y) + 2ab \operatorname{cov}(X, Y)$

Let X and Y be 2 independent CRVs with mean 0 and variance 1. It is given that

$$Z = 1 + X + XY^2$$
$$W = 1 + X$$

Find the cov(Z, W).

→ $\operatorname{cov}(Z, W) = \operatorname{cov}(1 + X + XY^2,\ 1 + X)$

$= \operatorname{cov}(X + XY^2,\ X)$ $\quad \because \operatorname{cov}(X + c, Y) = \operatorname{cov}(X, Y)$

$= \operatorname{cov}(X, X) + \operatorname{cov}(XY^2, X)$

$= \operatorname{var}(X) + \operatorname{cov}(XY^2, X)$

$= 1 + E[XY^2 - X] - E[XY^2] \cdot E[X]$

$= 1 + E[X^2Y^2] - E[XY^2] \cdot E[X]$

$= 1 + E[X^2] \cdot E[Y^2] - E[X] \cdot E[Y^2] \cdot E[X]$

$= 1 + 1 \cdot 1 - 0 \cdot 1 \cdot 0$

$= 2$

$\operatorname{Var}(X) = E[X^2] - E[X]^2$

$1 = E[X^2] - 0$

$E[X^2] = 1$

correlation :

Correlation of X and Y is normalizing the covariance. $\rho(X,Y)$
We define the standardized versions of X and Y as

$$u = \frac{X - E[X]}{\sigma_X} \qquad v = \frac{Y - E[X]}{\sigma_Y}$$

Then, correlation $(X,Y) = \rho(X,Y) = \text{cov}(u,v)$

$$= \text{cov}\left(\frac{X - E[X]}{\sigma_X}, \frac{Y - E[Y]}{\sigma_Y}\right)$$

$$= \text{cov}\left(\frac{X}{\sigma_X}, \frac{Y}{\sigma_Y}\right)$$

$$= \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y}$$

Uniform Random Variable $\mathcal{U}_{[a,b]}$

A random variable X is uniformly distributed over the interval [α, β] if its PDF is given by

$$f(x) = \begin{cases} \frac{1}{\alpha-\beta}, & \alpha \le x \le \beta \\ 0, & \text{otherwise} \end{cases}$$

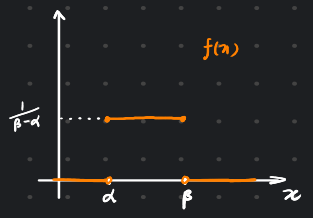
$$F_x(x) = \int_{-\infty}^{x} f_x(x)\,dx$$

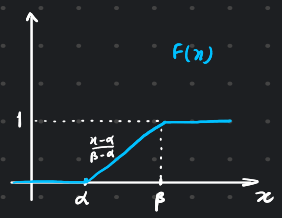
if $x < \alpha$, $F_x(x) = \int_{-\infty}^{x} f(x)\,dx = \int_{-\infty}^{x} 0 \cdot dx = 0$

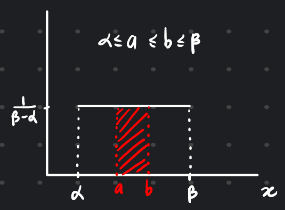
if $\alpha \le x \le \beta$, $F_x(x) = \int_{-\infty}^{x} f(x)\,dx = \int_{-\infty}^{\alpha} 0 \cdot dx + \int_{\alpha}^{x} \frac{1}{\beta-\alpha}\,dx = \left[\frac{x}{\beta-\alpha}\right]_{\alpha}^{x} = \frac{x-\alpha}{\beta-\alpha}$

if $x > \beta$, $F_x(x) = \int_{-\infty}^{\alpha} 0 \cdot dx + \int_{\alpha}^{\beta} \frac{1}{\beta-\alpha}\,dx + \int_{\beta}^{\infty} 0 \cdot dx = \frac{\beta-\alpha}{\beta-\alpha} = 1$

$$F_x(x) = \begin{cases} 0, & x < \alpha \\ \frac{x-\alpha}{\beta-\alpha}, & \alpha \le x \le \beta \\ 1, & x > \beta \end{cases}$$

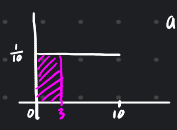




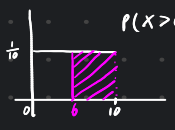


$$P\{a \le x \le b\} = \int_a^b f(x)\,dx = \int_a^b \frac{1}{\beta-\alpha}\,dx = \frac{b-a}{\beta-\alpha} = \frac{\text{Area}(\text{▨})}{\text{Area (total)}} = \frac{(b-a)\cdot\frac{1}{(\beta-\alpha)}}{(\beta-\alpha)\cdot\frac{1}{(\beta-\alpha)}} = \frac{b-a}{\beta-\alpha}$$

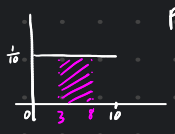
Q. $X \sim U_{[0,10]}$. Calculate a) $P(X<3)$ b) $P(X>6)$ c) $P(3<X<8)$



a) $P(X<3) = 3 \cdot \frac{1}{10} = \frac{3}{10}$

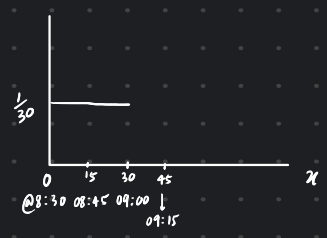


$P(X>6) = 4 \cdot \frac{1}{10} = \frac{2}{5}$



$P(3<X<8) = 5 \cdot \frac{1}{10} = 1/2$

Q. Buses arrive at a stop at 15 minute intervals starting at 07:00. If a passenger arrives at the stop at a time which is uniformly distributed between 08:30 and 09:00, find the probability that they have to wait
a. < 5 minutes b. > 10 minutes
for a bus.



x : # of mins. after 08:30 at which the person arrives

at exactly 8:30, bus goes.
So next bus is 8:45 or 9:00

$P(\text{waiting time} < 5 \text{ mins}) = P(\text{he comes bet}^{n}\ 08{:}40 - 08{:}45 \text{ or } 08{:}55 - 09{:}00)$

$= P(10 < x \le 15) + P(25 < x \le 30)$

$= 5 \cdot \frac{1}{30} + 5 \cdot \frac{1}{30} = 1/3$

$P(\text{waits} > 10 \text{ mins}) = P(08{:}30 - 08{:}35 \text{ or } 08{:}45 - 08{:}50)$

$= P(0 < x < 5 \text{ or } 15 < x < 20)$

$= 5 \cdot \frac{1}{30} + 5 \cdot \frac{1}{30} = 1/3$

Mean and variance of the uniform random variable

$X \sim U_{[\alpha, \beta]}$

$$f_X(x) = \begin{cases} \frac{1}{\beta-\alpha}, & \alpha \le x \le \beta \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x)\,dx = \int_{\alpha}^{\beta} x \frac{1}{\beta-\alpha}\,dx = \frac{1}{\beta-\alpha}\left[\frac{x^2}{2}\right]_{\alpha}^{\beta} = \frac{\beta^2-\alpha^2}{2(\beta-\alpha)} = \frac{(\alpha+\beta)}{2}$$

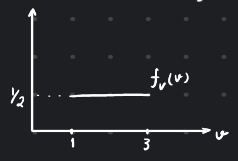
Ily, $$E[X^2] = \int_{\alpha}^{\beta} x^2 \cdot \frac{1}{\beta-\alpha}\,dx = \frac{1}{\beta-\alpha}\left[\frac{x^3}{3}\right]_{\alpha}^{\beta} = \frac{1}{3(\beta-\alpha)}(\beta^3-\alpha^3) = \frac{(\beta^2+\alpha^2+\alpha\beta)}{3}$$

$$\text{var}(X) = \frac{\alpha^2+\beta^2+\alpha\beta}{3} - \frac{\alpha^2+\beta^2+2\alpha\beta}{4} = \frac{\beta^2+\alpha^2-2\alpha\beta}{12} = \frac{(\beta-\alpha)^2}{12}$$

Q. The current in a semiconductor diode is often measured by the Shockley equation: $I = I_0(e^{aV} - 1)$, where V is the voltage across the diode, Io is the reverse current, a is a constant, and I is the resulting current in the diode.
Find the expected value of I, E[I], if a=5, Io = 10^-6 A, and V is uniformly distributed between [1, 3] volts.

→ $$E[I] = E[I_0(e^{aV} - 1)]$$
$$= I_0[E[e^{aV}] - 1]$$

$V \sim U_{[1,3]}$



$$E[e^{aV}] = \int_{-\infty}^{\infty} e^{aV} f_v(v)\,dv = \int_{1}^{3} e^{av} \frac{1}{2}\,dv = \frac{1}{2}\left[\frac{e^{av}}{a}\right]_{v=1}^{3}$$
$$= \frac{e^{3a} - e^{a}}{2a}$$

$$E[I] = I_0\left[\frac{e^{3a} - e^{a}}{2a} - 1\right]$$
$$= 10^{-6}\left[\frac{e^{15} - e^{5}}{10} - 1\right]$$
$$= (e^{15} - e^{5} - 10) \times 10^{-7}$$
$$= 0.32688$$

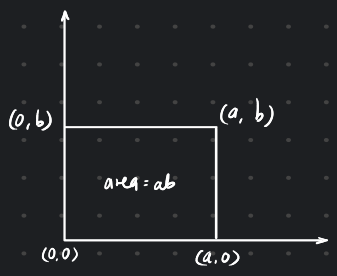
The random vector (X, Y) is said to be uniformly distributed over the 2-dimensional region R if its joint density function is constant for all points in R and zero for all other points.

$$f(x,y) = \begin{cases} c, & \text{if } (x,y) \in R \\ 0, & \text{otherwise} \end{cases}$$

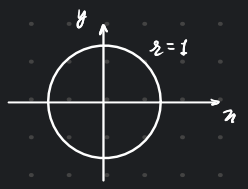
As $\iint_{x,y \in R} c \, dx\,dy = 1$, $\quad c = \dfrac{1}{\iint_{x,y \in R} dx\,dy} = \dfrac{1}{\text{Area of } R}$

e.g. X, Y is jointly uniformly distributed over the rectangular region 0<=x<=a, 0<=y<=b, then

$$f(x,y) = \begin{cases} \frac{1}{ab}, & 0 \le x \le a, \\ & 0 \le y \le b \\ 0, & \text{otherwise} \end{cases}$$



Q. Consider the unit disc: $D = \{(x,y) \mid x^2 + y^2 \le 1\}$
(Think of this as throwing darts on a dart board with a radius of 1 unit).
Suppose that we choose a random point (X, Y) uniformly at random in D.
The joint PDF of X and Y is given as



$$f_{xy}(x,y) = \begin{cases} c, & x^2+y^2 \le 1 \\ 0, & \text{else} \end{cases}$$

Find c.

→ $c = \dfrac{1}{\text{area}} = \dfrac{1}{\pi 1^2} = \dfrac{1}{\pi}$.

Find the marginal PDFs of X and Y.

→ $f_x(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$

$x^2 + y^2 \le 1$, $-1 \le x \le 1$. $\Rightarrow y^2 \le 1 - x^2$

$-\sqrt{1-x^2} \le y \le \sqrt{1-x^2}$

$$= \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi}\,dy$$

$$= \frac{2\sqrt{1-x^2}}{\pi}$$

$$f_x(x) = \begin{cases} \frac{2}{\pi}\sqrt{1-x^2}, & -1 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

By symmetry, $f_y(y) = \begin{cases} \frac{2}{\pi}\sqrt{1-y^2}, & -1 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$

Find the conditonal PDF of X given Y=y where -1 <= y <= -1.

$$f_{x|y}(x|y) = \frac{f_{xy}(x,y)}{f_y(y)}$$

if $-1 \le y \le 1$, $\quad f_{x|y}(x,y) = \dfrac{\frac{1}{\pi}}{\frac{2}{\pi}\sqrt{1-y^2}} = \dfrac{1}{2\sqrt{1-y^2}}$ $\quad y \ne 1$

also $-\sqrt{1-y^2} \le x \le \sqrt{1-y^2}$

$$f_{x|y}(x,y) = \begin{cases} \frac{1}{2\sqrt{1-y^2}}, & -\sqrt{1-y^2} \le x \le \sqrt{1-y^2} \\ 0, & \text{otherwise} \end{cases}$$

Are X and Y independent?

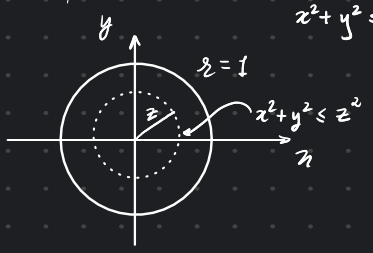
→ NO! $\quad f_{x,y}(x,y) \ne f_x(x) \cdot f_y(y)$.

"HW"

1) In the dart game considered above, instead of observing where the dart lands, we now observe how far the dart lands from the centre of the dart board. $Z \sim$ distance of dart from the centre

Find the pdf of Z, $E[Z]$, $\text{Var}(Z)$

→



$x^2 + y^2 \le 1$

$$F_Z(z) = P(Z \le z) = \frac{\text{Area}(x^2+y^2 \le z^2)}{\text{Area}(x^2+y^2 \le 1^2)} = \frac{\pi z^2}{\pi(1)} = z^2 \quad \text{if } 0 \le z \le 1$$

if $z < 0 \rightarrow P(Z \le z) = 0$

$z > 1 \rightarrow P(Z \le z) = 1$

$$F_Z(z) = \begin{cases} 0 & z < 0 \\ z^2 & z \in [0,1] \\ 1 & z > 1 \end{cases}$$

$$\therefore f_Z(z) = \begin{cases} 2z & z \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

$$E[Z] = \int_{-\infty}^{\infty} z \cdot f_Z(z)\,dz = \int_0^1 z \cdot 2z \cdot dz = 2\left[\frac{z^3}{3}\right]_{z=0}^1 = \frac{2}{3}$$

$$E[Z^2] = \int_0^1 z^2 \cdot 2z\,dz = 2\left[\frac{z^4}{4}\right]_0^1 = \frac{1}{2}$$

$$\text{Var}(Z) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

Q: Suppose Mr. and Mrs.Parab agree to meet at the Hanover inn between 05:00 and 06:00 p.m. on tuesday. Suppose each arrives at a time between 05:00 and 06:00 chosen at random with uniform probability. What is the distribution for the length of time that the first arrive has to wait for the other? What is the expected duration that the first arrive has to wait for the other?

→ let $X \sim U_{[0,1]}$, $Y \sim U_{[0,1]}$

X: # of hours after 05:00 p.m. that mr. Parab arrives.

Y: # of hours after 05:00 p.m. that mrs. Parab arrives.

ex. $X = 0.5$ means mr. Parab arrives at 05:30 p.m.

Z: # of hours the 1st arrive has to wait for the other.

$Z = |X - Y|$

X: $f_X(x) = \begin{cases} 1 & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$

same as $f_Y(y)$.



X and Y are independent.

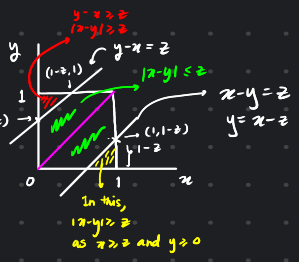
$f_{X,Y}(x,y) = \begin{cases} 1, & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$

$F_Z(z) = P(Z \le z)$

$= P(|X-Y| \le z)$

$= \frac{\text{Area}(\heartsuit)}{\text{Area}(\square)}$

$= \frac{1 - 2 \cdot \frac{1}{2} \cdot (1-z)(1-z)}{1}$



$F_Z(z) = 1 - (1-z)^2$

$F_Z(z) = \begin{cases} 0, & z < 0 \\ 1 - (1-z)^2, & z \in [0,1] \\ 1, & z > 1 \end{cases}$

$\Rightarrow f_Z(z) = \begin{cases} 2(1-z), & z \in [0,1] \\ 0, & \text{otherwise} \end{cases}$

$E[Z] = \int_{-\infty}^{\infty} z \cdot f_Z(z)\, dz = \int_0^1 z \cdot 2(1-z) = 2\left[\frac{z^2}{2} - \frac{z^3}{3}\right]_0^1 = 2\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{1}{3}$ hrs $= 20$ mins.

Q: Let X, Y both be $U_{(0,1)}$, $Z = XY$. Find PDF (Z). Find $E[Z]$.

→ $F_Z(z) = P(Z \le z)$

$= P(XY \le z)$

$= P\left(Y \le \frac{z}{X}\right)$ → as $y \in [0,1)$ and $\frac{z}{x}$ could be > 1

if $z \in [0,1]$,

$F_Z(z) = \int_0^1 \int_0^{\min(1, \frac{z}{x})} 1 \cdot dx\, dy = \int_0^1 \left[y\right]_0^{\min(1,\frac{z}{x})} dx = \int_0^1 \min\left(1, \frac{z}{x}\right) dx = \int_0^z 1\, dx + \int_z^1 \frac{z}{x}\, dx = z + z\left[\ln x\right]_z^1 = z - z\ln z$

$F_Z(z) = \begin{cases} 0, & z < 0 \\ z - z\ln z, & z \in [0,1] \\ 1, & z > 1 \end{cases}$

$\frac{d}{dz}(z - z\ln z) = 1 - \left(z \cdot \frac{1}{z} + \ln z \cdot 1\right) = -\ln z$

$f_Z(z) = \begin{cases} -\ln z, & z \in [0,1] \\ 0, & \text{otherwise} \end{cases}$

$\int u v\, dx = u \int v\, dx - \int \frac{du}{dx} \int v\, dx\, dx$

$E[Z] = \int_{-\infty}^{\infty} z \cdot f_Z(z)\, dz = \int_0^1 z(-\ln z) = -\int_0^1 z\ln z\, dz = -\left[\ln z \int z\, dz - \int \frac{1}{z}\left(\int z\, dz\right) dz\right] = -\left[\ln z \frac{z^2}{2} - \int \frac{z}{2}\, dz\right] = -\ln z \cdot \frac{z^2}{2} + \frac{z^2}{4} = \frac{z^2}{4} - \frac{z^2}{2}\ln z$

(v = $z\ln z$, u = dz)

$= \left[\frac{z^2}{4} - \frac{z^2}{2}\ln z\right]_0^1 = \frac{1}{4} - \frac{1}{2} \cdot 0 - 0 - 0 = \frac{1}{4}$

2) $X \sim$ Uniform $(0,1)$ — X and Y are independent
$Y \sim$ Uniform $(0,1)$ — Find the pdf of $X+Y$

→ $X, Y \sim U_{[0,1]}$ $\quad Z = X+Y \quad Z \in [0,2]$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z-x)\, dx$$

$$= \int_0^1 f_Y(z-x)\, dx$$

$$0 \le z-x \le 1$$
$$0 \ge x-z \ge -1$$
$$z \ge x \ge z-1$$

$0 \le z \le 1$ $\quad$ $1 \le z \le 2$

If $0 \le z \le 2$, $0 \le x \le z$

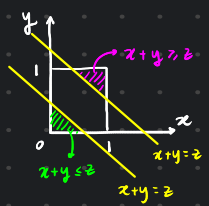
$$f_Z(z) = \int_0^z f_Y(x)\, dx = \int_0^z 1\, dx = z$$

If $1 \le z \le 2$, $z-1 \le x \le 1$

$$f_Z(z) = \int_{z-1}^1 1\, dx = 1-(z-1) = 2-z$$

$$f_Z(z) = \begin{cases} z, & 0 \le z \le 1 \\ 2-z, & 1 \le z \le 2 \\ 0, & \text{otherwise} \end{cases}$$

Method 2 : $f_{X,Y}(x,y) = \begin{cases} 1 & 0 \le x,y \le 1 \\ 0 & \text{otherwise} \end{cases}$



we'll calculate area differently for both the regions

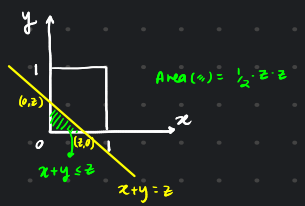
$Z = X+Y$

$F_Z(z) = P(Z \le z)$

$= P(X+Y \le z)$

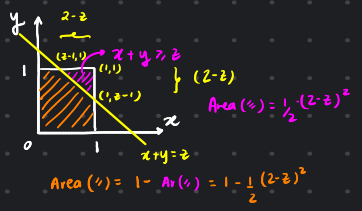
If $0 \le z \le 1$, $F_Z(z) = P(X+Y \le z)$

$$= \frac{z^2}{2}$$



If $1 \le z \le 2$, $F_Z(z) = P(X+Y \le z)$

$$= 1 - \frac{1}{2}(2-z)^2$$



$$\therefore F_Z(z) = \begin{cases} 0, & z < 0 \\ z^2/2, & z \in [0,1] \\ 1 - \frac{1}{2}(2-z)^2, & z \in [1,2] \\ 1, & z > 2 \end{cases}$$

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

$$= \begin{cases} z, & z \in [0,1] \\ (2-z), & z \in [1,2] \\ 0, & \text{otherwise} \end{cases}$$

$$E[Z] = \int_{-\infty}^{\infty} z \cdot f_Z(z)\, dz = \int_0^1 z \cdot z\, dz + \int_1^2 z(2-z)\, dz$$

$$= \left[\frac{z^3}{3}\right]_0^1 + \left[z^2\right]_1^2 + \left[\frac{z^3}{3}\right]_1^2$$

$$= \frac{1}{3} + 3 + \frac{1}{3} - \frac{8}{3}$$

$$= 1$$

Also $E[X+Y] = E[X] + E[Y]$

$= \frac{1}{2} + \frac{1}{2} = 1$

Also $\text{var}[X+Y] = \text{var}[X] + \text{var}[Y]$ as they're independent

Exponential Random Variable $X \sim Exp(\lambda)$

→ Decays exponentially

It is a CRV whose PDF is given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

where λ is a +ve constant.

CDF:

$$F_X(x) = \int_{-\infty}^{x} f_X(x)\,dx = \int_0^{\infty} \lambda e^{-\lambda x}\,dx$$

$$= \lambda \left[\frac{e^{-\lambda x}}{-\lambda}\right]_0^x$$

$$= \left[e^{-\lambda x}\right]_x^0$$

$$F_X(x) = 1 - e^{-\lambda x}$$

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x)\,dx$$

$$= \int_0^{\infty} x \cdot \lambda e^{-\lambda x}\,dx$$

$$= \lambda \int_0^{\infty} \underset{u}{x}\ \underset{v}{e^{-\lambda x}}\,dx \qquad uv - \int u' \int v$$

$$= \lambda \left[x\left[\frac{e^{-\lambda x}}{-\lambda}\right]_0^{\infty} - \int \frac{e^{-\lambda x}}{-\lambda}\,dx\right]_0^{\infty}$$

$$= \lambda \left[\frac{-xe^{-\lambda x}}{-\lambda} + \frac{1}{\lambda}\frac{e^{-\lambda x}}{-\lambda}\right]_0^{\infty}$$

$$= \left[xe^{-\lambda x} - \frac{e^{-\lambda x}}{\lambda}\right]_0^{\infty}$$

$$= 0 - 0 - \left[0 - \frac{1}{\lambda}\right] = \frac{1}{\lambda}$$

$\lim_{x\to\infty} \frac{x}{e^{\lambda x}} = \lim_{x\to\infty} \frac{1}{\lambda e^{\lambda x}} = 0$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\,dx$$

$$= \int_0^{\infty} x^2 \lambda e^{-\lambda x}\,dx$$

$$= \lambda \int_0^{\infty} \underset{u}{x^2}\ \underset{v}{e^{-\lambda x}}\,dx$$

$$= \lambda \left[x^2 \cdot \frac{e^{-\lambda x}}{-\lambda} - \int 2x \frac{e^{-\lambda x}}{-\lambda}\,dx\right]_0^{\infty}$$

$$= \left[-x^2 e^{-\lambda x}\right]_0^{\infty} + 2\int_0^{\infty} x e^{-\lambda x}\,dx$$

$$= [0-0] + 2\int_0^{\infty} x e^{-\lambda x}\,dx$$

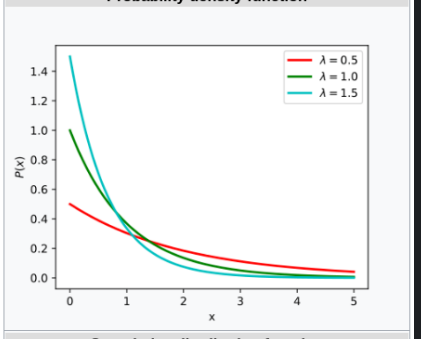
$$= \frac{2}{\lambda} \cdot \int_0^{\infty} x \lambda e^{-\lambda x}\,dx$$

$$= \frac{2}{\lambda} E[X] = \frac{2}{\lambda} \cdot \frac{1}{\lambda} = \frac{2}{\lambda^2}$$

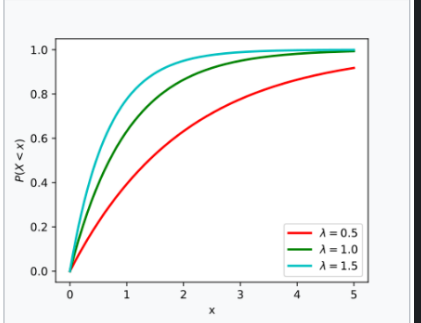
$$Var(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{2-1}{\lambda^2}$$

Exponential

Probability density function



Cumulative distribution function



Parameters	$\lambda > 0$, rate, or inverse scale
Support	$x \in [0, \infty)$
PDF	$\lambda e^{-\lambda x}$
CDF	$1 - e^{-\lambda x}$

$$E[X] = \frac{1}{\lambda}$$

$$Var(X) = \frac{1}{(1-\lambda)}$$

Q. Suppose that the length of a phone call in minutes is an exponential variable with parameter λ = 1/10. If someone arrives at immediately ahead of you at a public telephone booth. Find the probability that you'll have to wait a) more than 10 minutes; b) between 10 and 20 minutes.

→ $\lambda = 1/10$ length of phone call : x

$$f_X(x) = \frac{1}{10} e^{-\frac{x}{10}} \quad x \geq 0$$

$$F_X(x) = 1 - e^{-\frac{1}{10}x} \quad x \geq 0$$

a) $P(X \geq 10) = 1 - F_X(10)$

$$= 1 - \left[1 - e^{-\frac{1}{10}\cdot 10}\right]$$

$$= \frac{1}{e}$$

b) $P(10 < X < 20) = F_X(20) - F_X(10)$

$$= 1 - e^{-2} - \left[1 - e^{-1}\right]$$

$$= \frac{1}{e} - \frac{1}{e^2}$$

Q. Suppose that the # of kms a car can run before its battery runs out is an exponential distributed rv with average of 10,000 kms. If a person desires to take a 5,000 kms trip, what is the probability that she will be able to complete the trip without repeating replacing the battery?

→ $E[X] = 10$ (say unit is 1000 kms)

$$\frac{1}{\lambda} = 10$$

$$\lambda = \frac{1}{10}$$

$$P(X \geq 5) = 1 - F_X(5)$$

$$= 1 - \left[1 - e^{-\frac{1}{10}\cdot 5}\right]$$

$$= e^{-1/2} \approx 0.604$$

Exponential variables are memoryless.

Memoryless Random Variables

We say that a random variable is memoryless iff

$$P\{X > s+t \mid X > t\} = P\{X > s\} \qquad \forall\ s,t \geq 0$$

$$\text{or} \quad \frac{P\{X > s+t, X > t\}}{P\{X > t\}} = P\{X > s\}$$

$$\Rightarrow P\{X > s+t\} = P\{X > s\} \cdot P\{X > t\}$$

For the exponential random variable,

$$\begin{aligned} P\{X > s\} \cdot P\{X > t\} &= (1 - F_X(s)) \cdot (1 - F_X(t)) \\ &= e^{-\lambda s} e^{-\lambda t} \\ &= e^{-\lambda(s+t)} \\ &= 1 - F_X(s+t) \\ &= P\{X > s+t\} \end{aligned}$$

$\therefore$ memoryless

The uniform rv is not memoryless.

$$P\{X > s\} = 1 - P\{X < s\} = 1 - \frac{s-a}{b-a} \qquad P\{X > t\} = 1 - \frac{t-a}{b-a} \qquad \text{say} \quad a < s, t < b$$

$$P\{X > s\} \cdot P\{X > t\} = \left(1 - \frac{s-a}{b-a}\right)\left(1 - \frac{t-a}{b-a}\right) \neq \left(1 - \frac{(s+t)-a}{b-a}\right) \qquad \text{not.}$$

If X1, X2, ..., Xn are independent exponential random variables having respective parameters λ1, λ2, ..., λn,
then min(X1, X2, ..., Xn) is also an exponential random variable
with parameter λ1+λ2+...+λn.

Proof:

$$X_i = \text{Exp}(\lambda)$$

$$Y = \min(X_1, X_2, \ldots, X_n)$$

$$\begin{aligned} F_Y(y) &= P\{Y \leq y\} \\ &= P\{\min(X_1, \ldots, X_n) \leq y\} \\ &= 1 - P\{\min(X_1, \ldots X_n) > y\} \end{aligned}$$

$$\begin{aligned} P\{\min(X_1, \ldots, X_n) > y\} &= P\{X_1 > y, X_2 > y, \ldots, X_n > y\} \\ &= P\{X_1 > y\} \cdot P\{X_2 > y\} \cdots P\{X_n > y\} \\ &= (1 - F_{X_1}(y)) \cdot (1 - F_{X_2}(y)) \cdots (1 - F_{X_n}(y)) \\ &= e^{-\lambda_1 y} e^{-\lambda_2 y} \cdots e^{-\lambda_n y} \\ &= e^{-(\lambda_1 + \cdots + \lambda_n) y} \end{aligned}$$

$$F_Y(y) = 1 - e^{-(\lambda_1 + \cdots + \lambda_n) y}$$

$\therefore$ Y is an exp. r.v. with parameter $\sum_{i=1}^{n} \lambda_i$.

Q. A series system is one that needs all its components to function for the system itself to function. For a 3-component system, the lifetimes (in hrs) are independent exponential rvs with parameters λ1=1, λ2=2, λ3=3. What is the probability that the system survives for 10 minutes.

→ $X_1 \sim \text{Exp}(\lambda=1)$, $X_2 \sim \text{Exp}(\lambda=2)$, $X_3 \sim \text{Exp}(\lambda=3)$

↳ lifetime of part 1

$Y = \min(X_1, X_2, X_3) \sim \text{Exp}(\lambda = 1+2+3 = 6)$

↳ lifetime of system

$\therefore P(Y > 10 \text{ mins}) = P(Y > \frac{1}{6} \text{ hrs})$

$= 1 - P(Y \leq 1/6)$

$= e^{-\frac{1}{6} \cdot 6}$

$= e^{-1} = \frac{1}{e} = 36.8\%$

If X is an exponential rv with parameter λ, then cX is also an exponential rv with parameter λ/c.

Proof: $Y = cX$

$$F_Y(y) = P\{Y \le y\} = P\{cX \le y\} = P\{X \le \tfrac{y}{c}\} = 1 - e^{-\frac{\lambda}{c}y}$$

$$F_Y(y) = \begin{cases} 1 - e^{-\frac{\lambda}{c}y}, & y \ge 0 \\ 0, & y < 0 \end{cases}$$

$$\Rightarrow f_Y(y) = \begin{cases} \frac{\lambda}{c} e^{-\frac{\lambda}{c}y}, & y \ge 0 \\ 0, & \text{otherwise} \end{cases}$$

$$Y = cX \sim \text{Exp}\left(\tfrac{\lambda}{c}\right)$$

Q. Suppose that the # of customers arriving at a store obeys a Poisson distribution with an average of λ customers per unit time, i.e. if Y is the # of customers arriving in an interval of length t, then Y ~ Pois(λt). Suppose that the store opens at time t=0. Let X be the arrival time of the first customer. Find PDF(X).

→ $Y \sim \text{Pois}(\lambda t)$

$$f_Y(y) = \begin{cases} e^{-\lambda t} \frac{(\lambda t)^y}{y!} & y = 1, 2, \dots, \infty \end{cases}$$

X : arrival time of first customer.

$$F_X(x) = P\{X \le x\}$$

$$P\{X > x\} = P\{\underset{\text{time } t=x}{\text{at}} Y = 0\}$$

$$f_{Y_{t=x}}(y) = \begin{cases} \frac{e^{-\lambda x}(\lambda x)^y}{y!}, & y = 1, 2, 3, \dots \\ 0, & \text{otherwise} \end{cases}$$

$$P\{Y = 0\} = e^{-\lambda x} \frac{(\lambda x)^0}{0!} = e^{-\lambda x}$$

$P\{X > x\} = e^{-\lambda x}$ → prob. that 0 customers arrive in x minutes.

$$\Rightarrow F_X(x) = P\{X \le x\} = 1 - P\{X > x\}$$

$$F_X(x) = 1 - e^{-\lambda x}$$

$$\Rightarrow f_X(x) = \lambda e^{-\lambda x} \quad \Rightarrow X \sim \text{Exp}(\lambda)$$

Q. Let X be a crv with PDF given by $f_X(x) = \begin{cases} 1 & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$
Let Y = -ln(1-X). Find PDF(Y).

→

$Y = -\ln(1-X)$ $\quad x \to 0$ to 1, $y \to 0$ to ∞

$$F_Y(y) = P\{Y \le y\} = P\{-\ln(1-x) \le y\} = P\{1 - x \ge e^{-y}\} = P\{x \le 1 - e^{-y}\} = F_X(1 - e^{-y}) = 1 - e^{-y} \quad (X \sim U_{[0,1]};\ \therefore F_X(x) = x)$$

$$F_Y(y) = \begin{cases} 1 - e^{-y}, & y \ge 0 \\ 0, & \text{otherwise} \end{cases}$$

$$f_Y(y) = \begin{cases} e^{-y}, & y \ge 0 \\ 0, & \text{otherwise} \end{cases} \qquad Y \sim \text{Exp}(1)$$

Q. Suppose X is a uniform crv with parameters a=1 and b=2, and given X=x, Y is an exponential rv with parameter λ=x.

$$Y \mid X=x \sim \text{Exp}(\lambda = x)$$

Find the values of E[Y] and var[Y].

→

$X \sim U_{[1,2]} \Rightarrow f_X(x) = \begin{cases} 1, & 1 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$

$f_{Y|X=x}(y) = \begin{cases} x e^{-xy}, & y \ge 0 \\ 0, & \text{otherwise} \end{cases}$

$E[Y] = E\left[E[Y|X=x]\right]$

$= E\left[\frac{1}{X}\right] = \int_{-\infty}^{\infty} \frac{1}{x} \cdot f_X(x)\,dx$

$= \int_{-\infty}^{\infty} E[Y/X=x] \cdot f_X(x)\,dx$

$= \int_1^2 \frac{1}{x} \cdot 1\,dx$

$= [\ln x]_1^2 = \ln 2$

$E[Y^2] = E\left[E[Y^2|X=2]\right]$

$\hookrightarrow E[Y^2|X=x] = \int_{-\infty}^{\infty} y^2 \cdot f_{Y|X}(y|X=x)\,dy$

$= \int_0^{\infty} y^2 \cdot x e^{-xy}\,dy$

$= \cdots = 2/x^2$

$E[Y^2] = E\left[\frac{2}{x^2}\right]$

$= \int_1^2 \frac{2}{x^2} \cdot 1 \cdot dx$

$= 2\left[\frac{-1}{x}\right]_1^2 = -2\left[\frac{1}{2} - \frac{1}{1}\right] = 2\left[\frac{1}{2}\right] = 1$

$Var(Y) = E[Y^2] - E[Y]^2$

$var(Y) = 1 - (\ln 2)^2$

or $var(Y) = E[var(Y|X)] + var(E[Y|X])$

$= E\left[\frac{1}{x^2}\right] + var\left[\frac{1}{x}\right]$

$= 1 - (\ln 2)^2$

Sums of Independent Random Variables

If X and Y are 2 CRVs, then Z = X+Y is a CRV as well.

$$Z = X + Y$$

$$F_Z(z) = P(Z \le z)$$

$$= P(X+Y \le z)$$

$$= \iint_{x+y \le z} f_{X,Y}(x,y)\, dx\, dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x,y)\, dy\, dx$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) \cdot f_Y(y)\, dy\, dx \qquad \because X, Y \text{ are independent.}$$

$$= \int_{-\infty}^{\infty} f_X(x) \left[\int_{-\infty}^{z-x} f_Y(y)\, dy \right] dx$$

$$F_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot F_Y(z-x)\, dx$$

$$f_Z(z) = \frac{d}{dz} \int_{-\infty}^{\infty} f_X(x)\, F_Y(z-x)\, dx$$

$$= \int_{-\infty}^{\infty} f_X(x) \frac{d}{dz} F_Y(z-x)\, dx$$

$$= \int_{-\infty}^{\infty} f_X(x)\, f_Y(z-x)\, dx$$

convolution

$$\therefore \boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z-x)\, dx}$$

shift

Sum of 2 independent exponential rvs

If X and Y are 2 independent exponential rvs with parameter λ,

$X \sim Exp(\lambda)$, $Y \sim Exp(\lambda)$

$Z = X + Y$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z-x)\, dx$$

$x > 0$

$z - x > 0$

$\Rightarrow x \leq z$

$$= \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)}\, dx$$

$$= \int_0^z \lambda^2 e^{-\lambda z}\, dx$$

$$= \lambda^2 e^{-\lambda z} \int_0^z dz$$

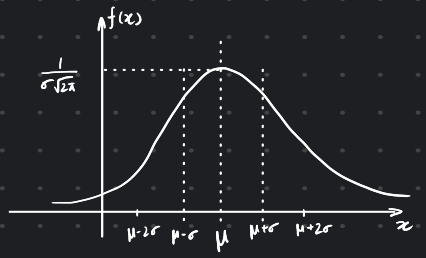
$$f_Z(z) = \lambda^2 e^{-\lambda z} z$$

Normal distribution $X \sim N(\mu, \sigma^2)$

A rv is said to be normally distributed with parameters μ and σ^2 if its density is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad x \in (-\infty, \infty)$$

f(x) is a bell-shaped curve symmetric about x=μ and attains a maximum value of $\frac{1}{\sigma\sqrt{2\pi}} = \frac{0.399}{\sigma}$ at x=μ.



An important implication of the preceding distribution is that if X is normally distributed with parameters μ and σ^2, then
Z = (X-μ)/σ is also normally distributed.

$$X \sim N(\mu, \sigma) \Rightarrow z = \frac{X-\mu}{\sigma} \sim N(0, 1)$$

It is the standard or the unit normal random variable.

PDF of standard normal random variable:

$X \sim N(\mu, \sigma^2)$ $\quad f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

$Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$

$$F_Z(z) = P\{Z \le z\}$$
$$= P\left\{\frac{X-\mu}{\sigma} \le z\right\}$$
$$= P\{X \le \sigma z + \mu\}$$
$$F_Z(z) = F_X(\sigma z + \mu)$$
$$f_Z(z) = \frac{d}{dz} F_Z(z) = \frac{d}{dz} F_X(\sigma z + \mu)$$
$$= \frac{d}{dz}(\sigma z + \mu) \cdot F_X'(\sigma z + \mu)$$
$$= \sigma \cdot f_X(\sigma z + \mu)$$
$$= \sigma \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\sigma z}{\sigma}\right)^2}$$

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \qquad z \in (-\infty, \infty)$$

Expectation of the standard normal distribution

$$E[Z] = \int_{-\infty}^{\infty} z \cdot f_Z(z)\, dz$$
$$= \int_{-\infty}^{\infty} z \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz$$
$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z\, e^{-z^2/2}\, dz$$
$$= \frac{1}{\sqrt{2\pi}} \left[-e^{-z^2/2}\right]_{-\infty}^{\infty}$$
$$= 0$$

$\int z \cdot e^{-z^2/2} dz$

let $-\frac{z^2}{2} = t$

$\Rightarrow -z\, dz = dt$

$\int z\, e^{-z^2/2} dz = \int -e^t dt = -e^t = -e^{-z^2/2}$

Variance of the standard normal distribution

$$\text{Var}(Z) = E[Z^2] - E[Z]^2$$

$$= \int_{-\infty}^{\infty} z^2 f_Z(z)\, dz - 0$$

$$= \int_{-\infty}^{\infty} z^2 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}}\, dz$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \left(\sqrt{2\pi} - z e^{-z^2/2}\right)_{-\infty}^{\infty}$$

$$= 1$$

$$\int z^2 e^{-z^2/2}\, dz$$

$$= \int z \cdot z e^{-z^2/2}\, dz$$

u ↪ we've already calculated the integral of this exp^n.

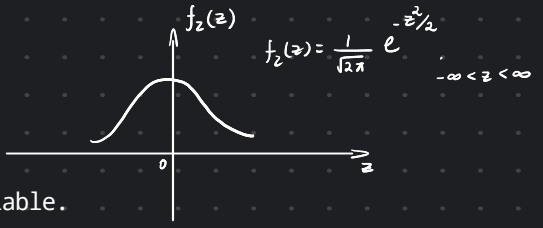
$$= z\left(-e^{-z^2/2}\right) - \int 1\left(-e^{-z^2/2}\right) dz$$

$$= -z e^{-z^2/2} + \sqrt{2\pi}$$

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} = 1 \Rightarrow \int_{-\infty}^{\infty} e^{-z^2/2} = \sqrt{2\pi}$$

$\text{Var}(Z) = \sigma^2 = 1$

$\Rightarrow$ std dev^n $(Z) = \sigma = 1$



The mean of the standard normal random variable is 0 and its variance is 1.

Using these, we can calculate the mean and variance of a normal random variable.

$$Z = \frac{X - \mu}{\sigma} \quad \Rightarrow X = \sigma Z + \mu$$

$$E[X] = E[\sigma Z + \mu] = \sigma E[Z] + \mu = \mu$$
$$\text{Var}(X) = \text{Var}(\sigma Z + \mu) = \sigma^2 \text{Var}(Z) = \sigma^2$$

If X is normally distributed with parameters μ and σ², then
Y = aX+b is also normally distributed with parameters aμ+b and a²σ².

Proof:

$$Y = aX + b$$

$$F_Y(y) = P\{Y \le y\}$$

$$= P\{aX + b \le y\}$$

$$= P\left\{X \le \frac{y-b}{a}\right\}$$

$$= F_X\left(\frac{y-b}{a}\right)$$

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

$$= \frac{d}{dy} F_X\left(\frac{y-b}{a}\right)$$

$$= \frac{d}{dy}\left(\frac{y-b}{a}\right) \cdot F_X'\left(\frac{y-b}{a}\right)$$

$$= \frac{1}{a} \cdot f_X\left(\frac{y-b}{a}\right) = \frac{1}{a} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\frac{y-b}{a} - \mu}{\sigma}\right)^2}$$

$$= \frac{1}{a\sigma\sqrt{2\pi}} e^{\frac{1}{2}\left(\frac{y-b-a\mu}{a\sigma}\right)^2}$$

$$\sim N\left(a\mu + b,\ a^2\sigma^2\right) \quad .$$

Cumulative distribution of the standard normal distribution

$X \sim Z$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \qquad x \in \mathbb{R}$$

$$F_X(x) = \int_{-\infty}^{x} f_X(y)\,dy = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\,dy$$

→ well defined function, given in the z-table

$$\phi(x) = F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\,dy$$

Area $\phi(z)$ under the standard normal curve to the left of z:

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Cumulative distribution of the normal distribution

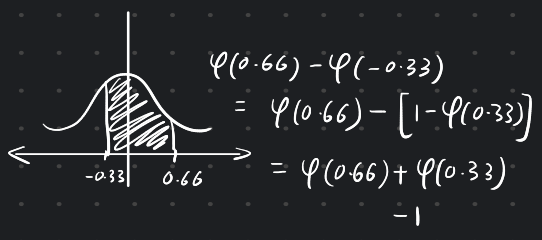
$$X \sim N(\mu, \sigma^2)$$

$$F_X(x) = P\{X \le x\}$$
$$= P\left\{\frac{X-\mu}{\sigma} \le \frac{x-\mu}{\sigma}\right\}$$
$$= P\left\{Z \le \frac{x-\mu}{\sigma}\right\}$$
$$= \varphi\left(\frac{x-\mu}{\sigma}\right)$$

Q. If X is a normal rv with mean 3 and variance 9, find:
a. P { 2 < X < 5 }
b. P { X > 0 }
c. P { |X-3| > 6 }

$\longrightarrow$ $\mu = 3, \quad \sigma^2 = 9$

$$P\{2 < X < 5\} = P\left\{\frac{2-3}{3} < Z < \frac{5-3}{3}\right\}$$
$$= P\{-0.33 < Z < 0.66\}$$
$$= \varphi(0.67) + \varphi(0.33) - 1$$
$$= 0.7468 + 0.6293 - 1$$
$$= 0.3779$$

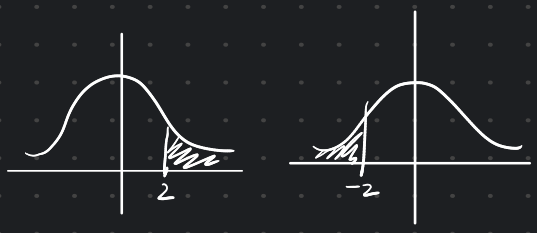


$$\varphi(0.66) - \varphi(-0.33)$$
$$= \varphi(0.66) - [1 - \varphi(0.33)]$$
$$= \varphi(0.66) + \varphi(0.33) - 1$$

$$P\{X > 0\} = P\{Z > -1\}$$
$$= \varphi(1)$$
$$= 0.8413$$



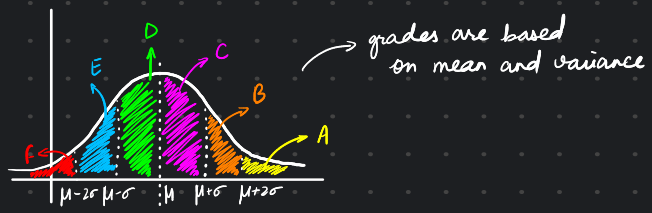
$$P\{|X-3| > 6\} = P\{X > 9 \text{ or } X < -3\}$$
$$= P\{X > 9\} + P\{X < -3\}$$
$$= P\{Z > 2\} + P\{Z < -2\}$$
$$= 2[1 - \varphi(2)]$$
$$= 0.0456$$



Applications of normal distributions

- Grading in exams:

  Often normal grading is used to give a measure of relative grade of students.



eg. $\phi(0.5) = 0.6915$
$\phi(0.25) = 0.5987$
where $\phi(x)$ is the cumulative density of standard normal distribution at x.

Let $X \sim N(2,4)$ $\mu = 2$, $\sigma^2 = 4$. and $Y = 3 - 2X$. Find:

a) $P(X > 1)$ b) $P(-2 < Y < 1)$ c) $P(X > 2 \mid Y < 1)$

→ $Y = 3 - 2X$

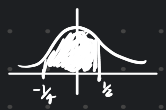
X : $\mu = 2$, $\sigma^2 = 4$, $\sigma = 2$

Y : $\mu = 3 - 2(2) = -1$, $\sigma^2 = 2^2 \cdot \sigma_X^2 = 16$, $\sigma = 4$

a) $P\{X > 1\} = P\{Z > \frac{-1}{2}\}$
$= \varphi(\frac{1}{2}) = 0.6915$



b) $P(-2 < Y < 1) = P\{\frac{-1}{4} < Z < \frac{1}{2}\}$
$= \varphi(\frac{1}{2}) + \varphi(\frac{1}{4}) - 1$
$= 0.6915 + 0.5987 - 1$
$= 0.29$



$\varphi(\frac{1}{2}) - \varphi(-\frac{1}{4})$
$= \varphi(\frac{1}{2}) - [1 - \varphi(\frac{1}{4})]$

c) $P\{X > 2 \mid Y < 1\} = \frac{P\{X > 2, Y < 1\}}{P\{Y < 1\}} = \frac{P\{X > 2, 3 - 2X < 1\}}{P\{3 - 2X < 1\}}$

$= \frac{P\{X > 2, X > 1\}}{P\{X > 1\}} = \frac{P\{X > 2\}}{P\{X > 1\}}$

$= \frac{P\{Z > 0\}}{P\{Z > 0.5\}} = \frac{0.5}{\varphi(0.5)}$



$= \frac{0.5}{0.6915} = 0.72$

eg. The power W dissipated in a resistor is proportional to the square of the voltage V. That is
$W = rV^2$, where r is a constant.

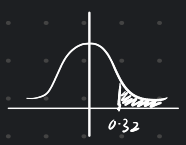
If $r = 3$, and V can be assumed as an approximation to be a normal random variable with mean 6 and S.D = 1, find a) $E[W]$ b) $P\{W > 120\}$

→ a) $W = 3V^2$ $\sim N(6,1)$

$E[W] = E[3V^2] = 3E[V^2]$
$= 3\left[\text{var}(V) + E[V]^2\right]$
$= 3[1 + 36] = 111$

b) $P\{3V^2 > 120\}$
$= P\{V^2 > 40\}$
$= P\{V > \sqrt{40}\}$
$= P\{Z > \frac{\sqrt{40} - 6}{1}\}$
$= P\{Z > 0.3245\}$
$= 1 - \varphi(0.32)$
$= 0.373$

... assuming voltage is the



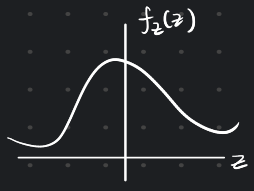
Q. Let $X \sim N(0, \sigma^2)$. Find $E[|X|]$.

$z = \frac{x-\mu}{\sigma}$

$X = \sigma z + \mu = \sigma z$

$\longrightarrow$ $Z \sim N(0,1)$

$E[|X|] = \sigma \cdot E[|Z|]$



$$E[|Z|] = 2\int_0^\infty z \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = \frac{2}{\sqrt{2\pi}} \int_0^\infty z\, e^{-z^2/2} dz = \frac{2}{\sqrt{2\pi}} \left[-e^{-z^2/2}\right]_0^\infty$$

$$= \sqrt{\frac{2}{\pi}} \left[0 + 1\right]$$

$$= \sqrt{\frac{2}{\pi}}$$

$$E[|X|] = \sigma\, E[|Z|] = \sigma\sqrt{\frac{2}{\pi}}$$

Let X and Y be 2 independent standard normal rvs and let Z = X+Y. Find the PDF of Z.

$$f_Z(z) = f_X(z) * f_Y(z)$$

$$= \int_{-\infty}^{\infty} f_X(x)\, f_Y(z-x)\, dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-\frac{(z-x)^2}{2}}\, dx$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[x^2+(z-x)^2\right]}\, dx$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[2x^2+z^2-2zx\right]}\, dx$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\cdot 2\left[(x-\frac{z}{2})^2+\frac{z^2}{4}\right]}\, dx$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(x-\frac{z}{2})^2-\frac{z^2}{4}}\, dx$$

$$= \frac{e^{-\frac{z^2}{4}}}{2\pi} \int_{-\infty}^{\infty} e^{-(x-\frac{z}{2})^2}\, dx$$

$$= \frac{e^{-z^2/4}}{2\pi} \cdot \sqrt{\pi}$$

$$= \frac{e^{-z^2/4}}{2\sqrt{\pi}}$$

$$f_Z(z) = \frac{1}{2\sqrt{\pi}} e^{-z^2/4}, \quad z \in (-\infty, \infty)$$

$$\sim N(\mu=0, \sigma^2=2)$$

$$2x^2+z^2-2zx$$
$$= 2\left[x^2-zx+\frac{z^2}{2}\right]$$
$$= 2\left[\left(x-\frac{z}{2}\right)^2+\frac{z^2}{4}\right]$$

we have a term $\int e^{-(x-\frac{z}{2})^2}\, dx$

we need to somehow make a normal variable out of this.

$$e^{\left[-\frac{1}{2}\frac{(x-\frac{z}{2})^2}{\frac{1}{2}}\right]}$$

$$= e^{-\frac{1}{2}\left(\frac{x-z/2}{1/\sqrt{2}}\right)^2}$$

this is in $N(z/2, (1/\sqrt{2})^2)$

$$\int_{-\infty}^{\infty} \frac{1}{1/\sqrt{2}\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-z/2}{1/\sqrt{2}}\right)^2} dx = 1$$

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}} e^{-(x-z/2)^2} dx = 1$$

$$\int_{-\infty}^{\infty} e^{-(x-z/2)^2} dx = \sqrt{\pi}$$

for $X, Y \sim N(0,1)$,

$X+Y \sim N(0,2)$

Theorem:

If $X \sim N(\mu_x, \sigma_x^2)$, $Y \sim (\mu_y, \sigma_y^2)$, then $X+Y \sim N(\mu_x+\mu_y, \sigma_x^2+\sigma_y^2)$. (if X and Y are independent)

Extension:

If $\{X_i\}_{i=1}^n \sim N(\mu_i, \sigma_i^2)_{i=1}^n$, then

$$Z = \sum_{i=1}^n X_i \sim N\left(\mu = \sum_{i=1}^n \mu_i, \sigma^2 = \sum_{i=1}^n \sigma_i^2\right)$$

Gamma distribution $\Gamma(\alpha, \lambda)$

A random variable is said to have a gamma distribution with parameters (α, λ), α, λ>0, if its density is given as

$$f_X(x) = \begin{cases} \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx$$

for any integer z, $\Gamma z = (z-1)!$

Γ is called the Gamma function.

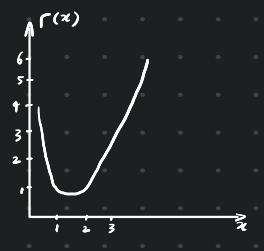
$$\Gamma(\alpha) = \int_0^\infty e^{-y} y^{\alpha-1}\, dy$$

$$\Gamma(1) = \int_0^\infty e^{-y}\, dy = 1$$

$$\Gamma(2) = \int_0^\infty e^{-y} y^1\, dy = 1$$

$$\Gamma(3) = \int_0^\infty e^{-y} y^2\, dy = 2$$

$$\Gamma(1/2) = \int_0^\infty e^{-y} y^{-1/2}\, dy = \sqrt{\pi}$$



$u\int v - \int u' \int v$

$$\Gamma(\alpha) = \int_0^\infty e^{-y} y^{\alpha-1}\, dy \quad (u = y^{\alpha-1},\ v = e^{-y})$$

$$= \left[y^{\alpha-1}\left(\frac{e^{-y}}{-1}\right) - \int (\alpha-1) y^{\alpha-2} \frac{e^{-y}}{(-1)}\, dy \right]_0^\infty$$

$$= \left[-e^{-y} y^{\alpha-1} \right]_0^\infty + (\alpha-1) \int_0^\infty y^{\alpha-2} e^{-y}\, dy$$

$$= 0 + (\alpha-1)\,\Gamma(\alpha-1)$$

$$\boxed{\Gamma(\alpha) = (\alpha-1)\,\Gamma(\alpha-1)}, \quad \Gamma(1) = 1,\ \Gamma(2) = 1$$

$$\Gamma(3) = 2\,\Gamma(2) = 2 \cdot 1$$

$$\Gamma(4) = 3\,\Gamma(3) = 3 \cdot 2 \cdot 1$$

$$\Gamma(5) = 4 \cdot \Gamma(4) = 4 \cdot 3 \cdot 2 \cdot 1$$

$$\Gamma(n) = (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1 = (n-1)!$$

when $\alpha = 1$, the gamma distribution reduces to exponential with mean $1/\lambda$.

Proof:

$$X \sim \text{Gamma}(\alpha=1, \lambda)$$

$$f_X(x) = \begin{cases} \frac{\lambda e^{-\lambda x}(\lambda x)^{1-1}}{\Gamma(1)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$= \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \sim \text{Exp}(\lambda)$$

→ Exp(λ) has mean $\frac{1}{\lambda}$

$$\therefore \text{Gamma}(1, \lambda) \sim \text{Exp}(\lambda)$$

Mean and variance of a gamma rv

$$f_X(x) = \begin{cases} \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$E[X] = \int_{-\infty}^\infty x \cdot f(x)\, dx$$

$$= \int_0^\infty x \cdot \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}\, dx = \frac{1}{\Gamma(\alpha)} \int_0^\infty e^{-\lambda x}(\lambda x)^\alpha\, dx$$

$$= \frac{1}{\lambda\,\Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda}(\lambda x)^\alpha\, dx$$

$$= \frac{1}{\lambda\,\Gamma(\alpha)}\,\Gamma(\alpha+1)$$

$$= \frac{1}{\lambda\,\Gamma(\alpha)} \cdot \alpha \cdot \Gamma(\alpha)$$

$$E[X] = \frac{\alpha}{\lambda}$$

$$\int_0^\infty \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)} = 1$$

$$\Rightarrow \Gamma(\alpha) = \int_0^\infty \lambda e^{-\lambda x}(\lambda x)^{\alpha-1}\, dx$$

$$E[X^2] = \int_0^\infty x^2 \cdot \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}\,dx$$

$$= \frac{1}{\lambda^2\Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x}(\lambda x)^{\alpha+1}\,dx$$

$$= \frac{1}{\lambda^2\Gamma(\alpha)} \cdot \Gamma(\alpha+2) = \frac{1}{\lambda^2\Gamma(\alpha)} \cdot (\alpha+1)(\alpha)\Gamma(\alpha) = \frac{\alpha^2+\alpha}{\lambda^2}$$

$$\text{Var}(X) = E[X^2] - E[X]^2$$

$$= \frac{\alpha^2+\alpha}{\lambda^2} - \frac{\alpha^2}{\lambda^2}$$

$$\text{Var}(X) = \frac{\alpha}{\lambda^2}$$

If $X \sim \text{Gamma}(\alpha, \lambda)$, then $E[X] = \frac{\alpha}{\lambda}$, $\text{Var}(X) = \frac{\alpha}{\lambda^2}$.

If $X_1, \dots, X_n$ are exponential r.v.s each with parameter λ,

then $\sum_{i=1}^n X_i$ is a Gamma r.v. with parameters (n, λ).

We've seen that $\text{Gamma}(1, \lambda) \sim \text{Exp}(\lambda)$.

Also sum of 2 independent Exp r.v.s with parameter λ has the PDF $f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z \geq 0 \\ 0, & \text{otherwise} \end{cases}$

$= \text{PDF of Gamma}(2, \lambda)$

and so on.

$$\text{Gamma}(\alpha, \lambda) \sim X_1 + X_2 + \dots + X_\alpha \qquad X_i \sim \text{Exp}(\lambda)$$

---

so if we add 2 independent Gamma r.v.s

$X_1 \sim \text{Gamma}(\alpha_1, \lambda)$, $X_2 \sim \text{Gamma}(\alpha_2, \lambda)$,

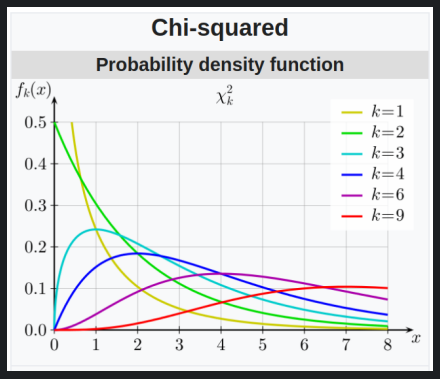
$X_1 + X_2 \sim \text{Gamma}(\alpha_1+\alpha_2, \lambda)$ as X_1 = Sum of α_1 independent Exp r.v.s (λ)

X_2 = sum of α_2 independent Exp r.v.s (λ)

$X_1 + X_2$ = Sum of $(\alpha_1+\alpha_2)$ independent Exp(λ) r.v.s

$= \text{Gamma}(\alpha_1+\alpha_2, \lambda)$

Chi-squared distribution



If Z1, Z2, ..., Zn are independent standard normal random variables, then X, defined as

$$X = Z_1^2 + Z_2^2 + \cdots + Z_n^2$$

is said to follow a Chi-squared distribution with n degrees of freedom.

$$X \sim \chi^2_n$$

The PDF of a Chi-squared distribution is given as

$$f_X(x) = \begin{cases} \dfrac{\frac{1}{2}\, e^{-x/2} \left(\frac{x}{2}\right)^{n/2-1}}{\Gamma(n/2)}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

A Chi-squared distribution with n degrees of freedom is same as the Gamma distribution with parameters (n/2, 1/2).

$$\chi^2_n = \Gamma\left(\frac{n}{2}, \frac{1}{2}\right)$$

PDF of Γ-distn (λ, α) :

$$\begin{cases} \dfrac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

The Chi-squared distribution has the additive property as well.
Adding 2 independent Chi-squared variables with parameters n1 and n2 would result in a Chi-squared distribution with (n1+n2) degrees of freedom.

Proof: $X_1 = Z_1^2 + \cdots + Z_{n_1}^2$, $X_2 = Z_1'^2 + Z_2'^2 + \cdots + Z_{n_2}'^2$ Z_is, Z_i's are std. normal variables.

$$X_1 + X_2 = Z_1^2 + \cdots + Z_{n_1}^2 + Z_1'^2 + \cdots + Z_{n_2}'^2$$
= sum of squares of (n_1+n_2) independent std. normal variables.
$$= \chi^2_{n_1+n_2}$$

Prove that $\chi^2_n = Z_1^2 + \cdots + Z_n^2$ where Z_i's are independent standard normal variables is equivalent to $\Gamma\left(\frac{n}{2}, \frac{1}{2}\right)$.

→ We'll first prove that $Z_1^2 \sim \Gamma\left(\frac{1}{2}, \frac{1}{2}\right)$, then add all of them up.

$Z_1 \sim N(0, 1)$ ... std. normal

PDF of Z_1: $f_{Z_1}(z) = \dfrac{1}{\sqrt{2\pi}} e^{-z^2/2}$, $z \in \mathbb{R}$

$Y = Z_1^2$. $F_Y(y) = P\{Y \leq y\}$

$$= P\{Z_1^2 \leq y\} = P\{-\sqrt{y} \leq Z_1^2 \leq \sqrt{y}\}$$
$$= F_{Z_1}(\sqrt{y}) - F_{Z_1}(-\sqrt{y})$$
$$= \varphi(\sqrt{y}) - \varphi(-\sqrt{y})$$

$F_Y(y) = \varphi(\sqrt{y}) - \varphi(-\sqrt{y})$

$f_Y(y) = \dfrac{d}{dy} F_Y(y)$

$$= \frac{1}{2\sqrt{y}} \left[f_{Z_1}(\sqrt{y}) + f_{Z_1}(-\sqrt{y})\right]$$
$$= \frac{1}{2\sqrt{y}} \cdot 2 f_{Z_1}(\sqrt{y}) = \frac{f_{Z_1}(\sqrt{y})}{(\sqrt{y})}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} \frac{e^{-y/2}}{\sqrt{y}} \sim \Gamma\left(\frac{1}{2}, \frac{1}{2}\right)$$

→ for $\Gamma\left(\frac{1}{2}, \frac{1}{2}\right)$,

$$f_Y(y) = \frac{\lambda e^{-\lambda y} (\lambda y)^{\alpha-1}}{\Gamma(\alpha)}$$
$$= \frac{\frac{1}{2} e^{-y/2} \left(\frac{1}{2} y\right)^{-1/2}}{\sqrt{\pi}} \qquad \Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}$$
$$= \frac{\left(\frac{1}{2}\right)^{1/2} e^{-y/2}}{\sqrt{\pi}} \frac{1}{\sqrt{y}} = \frac{1}{\sqrt{2\pi}} \frac{e^{-y/2}}{\sqrt{y}}$$

Both are equal.

That's another imp property of Gamma rvs:

For a std. normal variable Z, $Z^2 \sim \Gamma(\frac{1}{2}, \frac{1}{2})$.

Proof contd:

$$Z_1^2 + Z_2^2 + \cdots + Z_n^2 = \text{Gamma}\left(\frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2}, \frac{1}{2}\right)$$

$$= \Gamma\left(\frac{1}{2} \cdot n, \frac{1}{2}\right)$$

$$\chi_n^2 \sim \Gamma\left(\frac{n}{2}, \frac{1}{2}\right)$$

Γ r.v.s are additive.
α's add up.
λ's don't add up.

Expected value and variance for the Chi-squared distribution with n degrees of freedom

$X \sim \Gamma(\frac{n}{2}, \frac{1}{2})$

for Γ-dist^n, $E[X] = \frac{\alpha}{\lambda} = \frac{n/2}{1/2} = n$

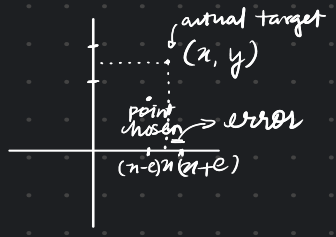
$$Var[X] = \frac{\alpha}{\lambda^2} = \frac{n/2}{1/4} = 2n$$

$E[X] = n$

$Var(X) = 2n$

Q. When we attempt to locate a target in a 2-D space, suppose that the coordinate errors are independent normal rvs with mean 0 and variance 2. Find the probability that the distance between the points chosen and the target exceeds 3.

→

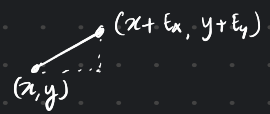


$e \sim N(0, 2)$

$P\{\text{distance} > 3\}$

$= P\{E_x^2 + E_y^2 > 9\}$

distance betn chosen points and target $= \sqrt{E_x^2 + E_y^2}$



Both $E_x, E_y \sim N(0, 2)$ and are independent.

$\frac{E_x - 0}{\sqrt{2}}, \frac{E_y - 0}{\sqrt{2}} \sim$ std. normal $N(0, 1)$

or you could think of E_x as $\sqrt{2} \cdot Z$

same for E_y

$$P\{\text{dist} > 3\} = P\{(\sqrt{2}Z_1)^2 + (\sqrt{2}Z_2)^2 > 9\}$$

$$= P\{Z_1^2 + Z_2^2 > 9/4\}$$

$$= P\{\chi_2^2 > 9/4\}$$

$$= P\{\Gamma(1, \tfrac{1}{2}) > 9/4\}$$

$$= P\{\text{Exp}(\tfrac{1}{2}) > 9/4\} = 1 - F_X(\tfrac{9}{4})$$

And for an Exp. r.v., $F_X(x) = 1 - e^{-\lambda x}$

$$P\{\text{dist} > 3\} = e^{-1/2 \cdot 9/4} = e^{-9/8} \approx 0.325$$

Student's t-distribution

If z and χ_n^2 are independent r.v.s where $z \sim N(0,1)$, χ_n^2 is the chi-squared dist^n with n degrees of freedom

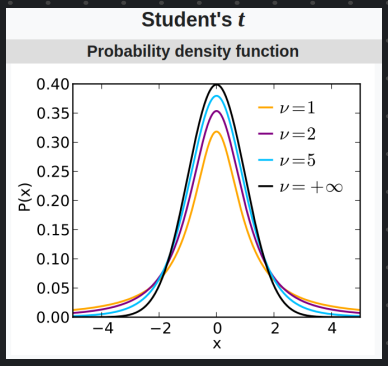
then $$t_n = \frac{z}{\sqrt{\chi_n^2/n}}$$

is said to be a t-distribution with n degrees of freedom.

For a very large n, $t_n = z$.

$$t_n = \frac{z}{\sqrt{z_1^2 + z_2^2 + \dots + z_n^2}}$$ ... z_i's are std. normal

$\sqrt{z_1^2 + z_2^2 + \dots + z_n^2}$ → RMS of std. normals



$E[T_n] = 0$

$\text{Var}(T_n) = \frac{n}{(n-2)}$, $n > 2$

For $n \to \infty$, $\frac{n}{n-2} \to 1$ (std. normal)